The two geographies of urban deprivation: everyday services and emergency care in 67 European cities

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Abstract

Cities concentrate people and the infrastructures that sustain them; a growing share of humanity depends on how well urban systems deliver the services of daily life and the emergency care on which survival depends. Urban theory treats access to such services as a quantity that improves predictably with city size, and measures it in travel minutes applied uniformly across needs. Yet the welfare burden of poor access is need-specific: it saturates for substitutable everyday services and escalates without bound for time-critical care. Here we measure potential deprivation rather than access, transferring calibrated welfare-loss functions from humanitarian logistics to a 100 m population grid covering 67 European functional urban areas in 24 countries: a bounded deprivation level for everyday services anchored on the 15-minute-city threshold, and an unbounded deprivation cost for emergency care anchored on clinical response benchmarks. The two regimes obey different laws. City size buys everyday access while emergency deprivation shows no detectable size gradient, and everyday inequality rises with size while emergency inequality never falls. Where the regimes diverge is structured by country and macro-region rather than by size, and the divergence culminates in a severity ordering of emergency-periphery coverage whose extreme, five capital-city emergency deserts, is invisible to access averages. The same residents are deprived in both regimes above chance in 66 of 67 cities, most strongly in Central-Eastern Europe, and children carry the compounded burden in 88 percent of cities. Urban growth delivers welfare regime by regime: everyday welfare through agglomeration, emergency welfare only through coverage.

Significance statement. Urban scaling theory holds that infrastructure and service provision improve systematically with city size, and accessibility research measures that provision in travel minutes. We show that the scaling regularity governs welfare for only one class of needs. Measuring anchored deprivation rather than travel time across 67 European city regions, everyday welfare improves with size while emergency welfare does not, and inequality in everyday welfare grows with size. Five European capitals harbour emergency deserts that minutes-based statistics cannot detect, and children are the most consistent carriers of compounded deprivation. Welfare-anchored deprivation rather than uniform travel time is the quantity urban theory should scale, with everyday and emergency capability entering that theory as separate regimes.

Keywords: urban scaling; deprivation cost; accessibility; 15-minute city; emergency medical services; spatial equity; resilience

1 Introduction

Urban critical infrastructure serves residents on two clocks. On the slow clock, people choose when to reach a general practitioner, a pharmacy, a supermarket, a school or a park; these trips recur often, and poor access to them can be coped with. On the fast clock, an ambulance or an emergency department must reach a patient, and minutes translate into survival probabilities. Whether urban growth serves both clocks, and who is left behind when it serves only one, is a question that sits at the intersection of three literatures, each of which supplies part of the answer and none of which can currently pose the question in welfare terms.

The first is urban scaling theory. Since the discovery that diverse urban quantities follow power laws of population with exponents falling into universality classes, with socio-economic outputs superlinear and infrastructure sublinear, scaling theory has treated the returns to city size as systematic and general [1]. Recent work has begun to condition that generality, showing for instance that the growth advantage of large cities depends on the urbanisation stage of their national system [2]. For service provision the scaling account makes a strong implicit prediction: if the returns to scale that concentrate facilities in larger cities are general, then access to everyday services and access to emergency care should improve together with city size. Whether that prediction holds has never been tested, because the scaling literature measures quantities of infrastructure rather than the welfare its spatial distribution delivers, and because it treats all service classes as one.

The second is the accessibility literature. Proximity-based planning doctrine, most visibly the 15-minute-city programme [3, 4], global minutes-based accessibility mapping [5, 6] and the measurement of unequal access within cities [7] have made travel time the common currency of urban service analysis. That currency embeds a measurement assumption so familiar it is rarely stated: a minute counts the same wherever it falls, for whichever service, at whichever point of the distribution. The assumption is false in both directions at once. For a substitutable everyday service, the difference between a 25- and a 45-minute walk is largely immaterial, because households reorganise routines around poor access; for emergency care, the difference between an 8- and a 15-minute response is clinically decisive. A uniform minutes scale therefore overstates welfare variation where it does not matter and understates it where it matters most, and any average over such a scale inherits both errors.

The third literature supplies the correction. Humanitarian logistics has formalised *deprivation*: the welfare cost of time spent without access to a good or service, estimable as a function of deprivation time [8, 9]. Two objects emerge from that programme, and their difference carries this paper. Deprivation from substitutable needs is a *level*: bounded and S-shaped, saturating once coping sets in [10]. Deprivation from critical needs is a *cost*: convex and unbounded, rising ever more steeply with time, with discrete-choice and stated-preference estimates available for goods and specifically for emergency ambulance services [11–13]. We transfer these functional forms to intra-urban access and calibrate their curvature to domain anchors: the 15-minute-city threshold for the everyday level, and the emergency-medical-service (EMS) response benchmarks for the emergency cost, under which deprivation is negligible below the 3–4 minute ideal, accumulates through the widely used and empirically contested 8-minute urban response standard, and reaches the benchmark cost at the 10–15 minute upper target [14–18]. Throughout the paper, *deprivation level* refers to the bounded everyday object (1 = full deprivation) and *deprivation cost* to the unbounded emergency object, reported in multiples of the cost at the 15-minute benchmark; the two are different mathematical objects by design, and only unit-invariant statistics compare them.

Treating the two regimes as distinct capabilities is also what resilience theory demands. Resilience has been defined through equitable access to essential services [19, 20], equality of access and network resilience have been shown to move together [21], and everyday vulnerability profiles have been shown empirically to fail as predictors of crisis-time capability [22]. If everyday functioning and emergency capability are separate dimensions, a city may deliver one and fail the other, and the population for whom both fail carries a *compounded* burden that no single-regime statistic reveals. What has been missing is a comparative measurement of both regimes, in welfare terms, across a continental urban system.

This paper provides that measurement. We compute the two deprivation surfaces on a common 100 m population grid for 67 functional urban areas (FUAs) in 24 European countries, from 101,050 to 12.9 million inhabitants, drawn by a stratified design over four macro-regions and four size strata. Following the scaling tradition we estimate log-log elasticities of city outcomes on population [1] under a harmonised city definition [2]. One qualifier applies to everything that follows and is stated here once: every size gradient in this paper is cross-sectional, and any phrase such as "as cities grow" means the space-for-time reading of that gradient, never a longitudinal observation.

Four questions organise the analysis, each aimed at one of the literatures above. RQ1: do larger cities provide better everyday services and better emergency capability, in levels and in inequality, as scaling generality would predict? RQ2: where do the two deprivations diverge, and what structures the divergence: size, country, region, or coverage? RQ3: who carries compounding deprivation, in terms of co-location typology and vulnerability strata? RQ4: what does analysing deprivation add over the access metrics that currently carry the accessibility literature?

The answers, previewed: agglomeration benefits are regime-specific (size buys everyday access and no detectable emergency relief, while everyday inequality rises with size); the divergence is structured by country and macro-region rather than size, and resolves into a one-dimensional severity ordering of emergency-periphery coverage whose extreme is five capital-city emergency deserts; compounding exceeds statistical independence almost everywhere and falls most consistently on children; and the deprivation layer changes no rank-based conclusion while halving unexplained variance and exposing the tail phenomena that access averages cannot represent. A robustness architecture, combining rank-invariance by construction, a specification curve, rank-agreement statistics, a routing-engine cross-check and an access-based re-estimation of every headline, is reported as a reusable template.

2 Results

What is measured

Two surfaces summarise each city; their construction is in Materials and Methods, and Fig. 1 shows the welfare logic that distinguishes them. The *everyday deprivation level* of a grid cell, between 0 and 1, expresses how far the burden of reaching five everyday service categories on foot has progressed toward saturation: near 0, all services are comfortably walkable; near 1, the cell is effectively without walkable everyday services and additional minutes no longer add burden. The *emergency deprivation cost* of a cell is an open-ended positive number expressing the welfare cost of its drive time to the nearest emergency facility, in multiples of the cost at the 15-minute EMS benchmark: a cell at 2.0 carries twice the deprivation cost of one whose ambulance arrives at the benchmark, and the cost keeps escalating beyond it. Both curves are anchored on external standards rather than fitted to the data, and the linear loss that any minutes average implicitly applies is drawn beside them in Fig. 1: relative to the calibrated curves, the linear loss over-prices minutes where they are welfare-inert (the saturated everyday range) and under-prices them where they are clinically decisive (around the emergency benchmarks).

The bounded level and the unbounded cost are deliberately different objects, because the needs are: everyday services are substitutable, so their deprivation saturates; emergency care is time-critical, so its deprivation escalates. Forcing both into levels would assert that a 60-minute ambulance is no worse than a 30-minute one; forcing both into costs would assert that living far from a supermarket is unboundedly catastrophic. The asymmetry is the substantive claim about

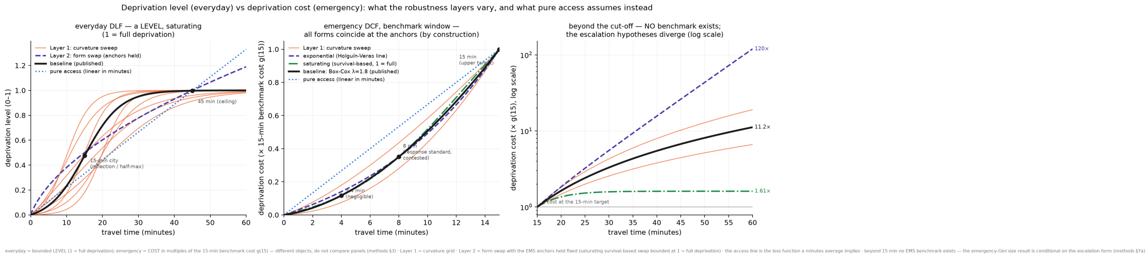

Figure 1: What matters, by regime: the two calibrated deprivation functions and the linear loss a minutes average implies. Left: everyday deprivation level (logistic, inflection at the 15-minute anchor; Layer-1 curvature sweep; concave Box-Cox form swap dashed), on its own 0–1 scale. Centre: emergency deprivation cost (convex Box-Cox, $\lambda = 1.8$, in multiples of g(15)) inside the benchmark window, where all forms coincide at the EMS anchors by construction (negligible at 4 min; ≈0.35 at the contested 8-min standard; 1 at the 15-min target). Right: beyond the 15-minute cut-off no benchmark exists and the three literature-grounded escalation hypotheses diverge (at 60 min: 1.61× saturating, 11.2× polynomial baseline, 120× exponential; log scale). The dotted line in each panel is the linear loss of a pure-access average. Each panel stays on its own reporting scale; panels are never compared in magnitude.

Table 1: Sample composition and median city indicators by macro-region. 67 functional urban areas (Eurostat/OECD definition: city plus commuting zone), stratified draw over 4 macro-regions × 4 size strata (10^5–$2.5 \cdot 10^5$, $2.5 \cdot 10^5$–10^6, 10^6–$5 \cdot 10^6$, $> 5 \cdot 10^6$ inhabitants; 15/18/30/4 cities), 113.9 million residents in total. $\bar{D}_{\mathrm{ev}}$: population-weighted mean everyday deprivation level (0–1); $\bar{C}_{\mathrm{em}}$: mean emergency deprivation cost (multiples of g(15); not comparable to $\bar{D}_{\mathrm{ev}}$ in magnitude); G: within-city population-weighted Gini; ρ: Spearman rank coupling of the two surfaces; HH: compounding population share at the median split. Source: `cities_descriptives.csv`.

Region	Cities	Countries	Pop. (M)	$\bar{D}_{\mathrm{ev}}$	$\bar{C}_{\mathrm{em}}$	G_{ev}	G_{em}	ρ	HH
North	13	4	11.3	0.41	1.81	0.39	0.47	0.47	0.34
West	16	5	37.1	0.23	0.74	0.52	0.41	0.36	0.32
South	18	4	39.8	0.29	1.07	0.42	0.42	0.38	0.32
CEE	20	11	25.6	0.30	1.48	0.51	0.54	0.55	0.36
All	67	24	113.9	0.29	1.14	0.49	0.44	0.42	0.33

what matters, and it imposes one discipline used throughout: only unit-invariant statistics cross regimes (elasticities, Ginis, rank-based measures), and raw magnitudes never do.

Table 1 summarises the sample; Table S3 in the SI lists every city. Fig. 2 gives the overview of all 67 cities: each city is a point in the plane spanned by its within-city inequality (Gini) of everyday and of emergency deprivation, with whiskers spanning the min–max envelope over the deprivation-curvature sensitivity grid, so each city is placed at the precision the calibration actually supports. Because cities nest in 24 countries, every scaling claim below cites a wild cluster bootstrap p-value on country clusters and every regional contrast a country-permutation p-value; the inference conventions are in Materials and Methods.

2.1 City size buys everyday access; emergency relief shows no detectable size gradient

If the returns to scale behind urban service provision were general, as the universality claims of scaling theory suggest for infrastructure [1], both deprivations should fall with city size. The everyday regime behaves exactly as that account predicts. Mean everyday deprivation falls systematically with population (Fig. 3): the log-log elasticity is −0.196 (country-clustered SE 0.023, $p_{\mathrm{wild}} = 0.0002$), meaning mean everyday deprivation falls by roughly 19 % for each e-fold (≈ 2.7×) of population, and city size alone explains 44 % of the cross-city variance ($R^2 = 0.44$). The mechanism is visible in the descriptive layer beneath the deprivation surfaces: the

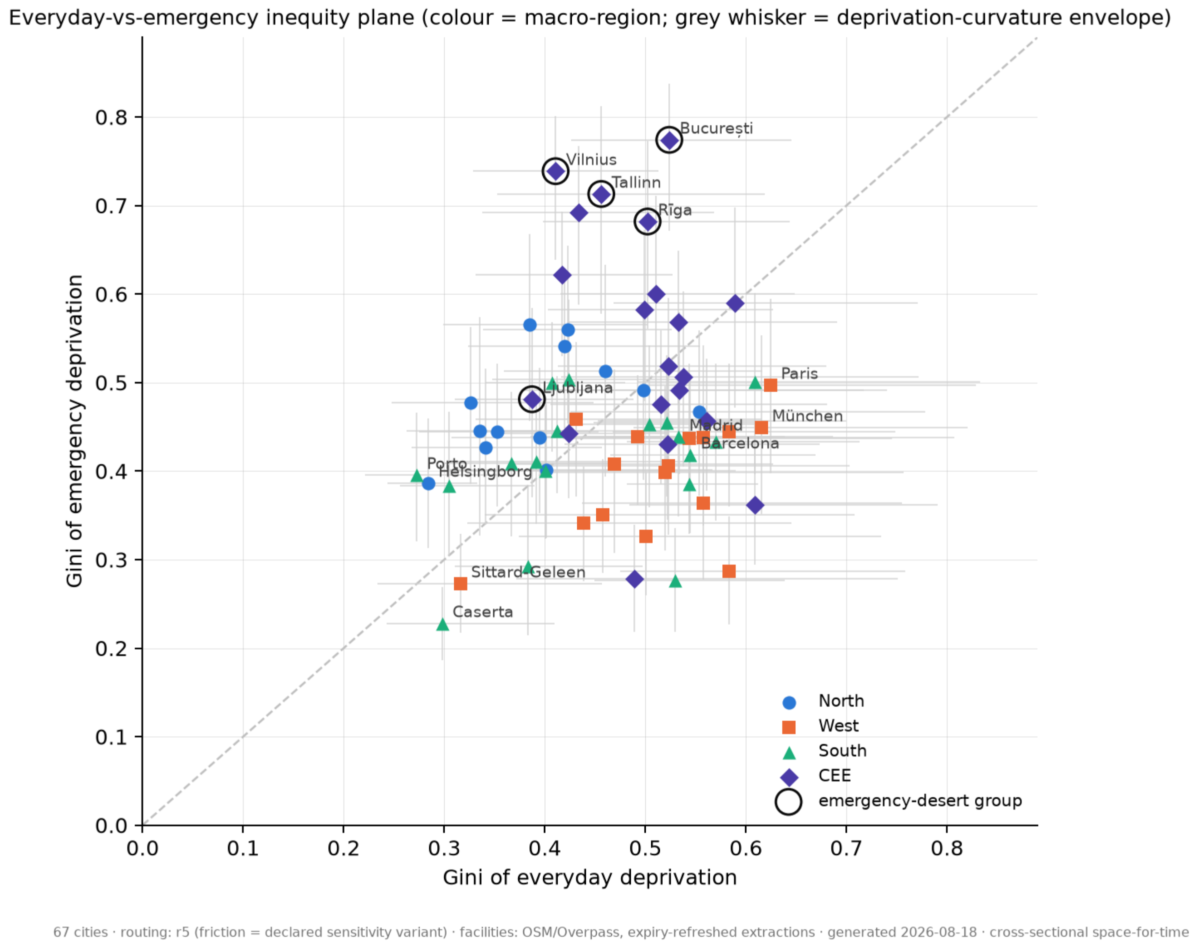

Figure 2: All 67 cities in the inequality plane. Horizontal axis: within-city Gini of the everyday deprivation level; vertical axis: within-city Gini of the emergency deprivation cost. Whiskers span the min–max envelope of each Gini over the deprivation-curvature sensitivity grid (Section 3.1). Colours mark macro-regions; the emergency-desert group of Section 2.3 is outlined.

median FUA routes its population to 93 mapped general practitioners, 228 pharmacies, 325 supermarkets, 409 schools and 636 green spaces, and the population-weighted median walk to the nearest facility of each everyday category is 4–8 minutes; larger cities carry denser facility networks, and the deprivation level converts that density into welfare terms.

The estimate survives every diagnostic we throw at it. It is stable under country fixed effects (-0.17), under leave-one-out city deletion (envelope -0.204 to -0.185, with Athina the most influential city), under outlier-robust estimators (Huber -0.196, median regression -0.223) and under exclusion of the five emergency-desert capitals (-0.203), so no single city, country or estimator choice carries it.

The emergency regime breaks the pattern. Mean emergency deprivation cost shows no detectable size gradient: the elasticity is -0.059 (clustered $p = 0.16$, $p_{\text{wild}} = 0.21$), and city size explains 2 % of the cross-city variance. Non-significance alone would be weak evidence, so the null is bounded: two one-sided equivalence tests place the emergency elasticity within ± 0.13 [23]. The bound licenses a precise statement and no more: any emergency size gradient, if present, is at most two-thirds the everyday one in magnitude; equivalence at the stricter ± 0.10 bound is not shown, so the claim is "no detectable scaling, bounded within ± 0.13", never "proven flat". The descriptive layer again shows why: the median FUA has 6 emergency departments and 6 ambulance stations for its whole territory, the median population-weighted drive to either is 12 minutes, and the facility count of a national emergency system does not multiply with city

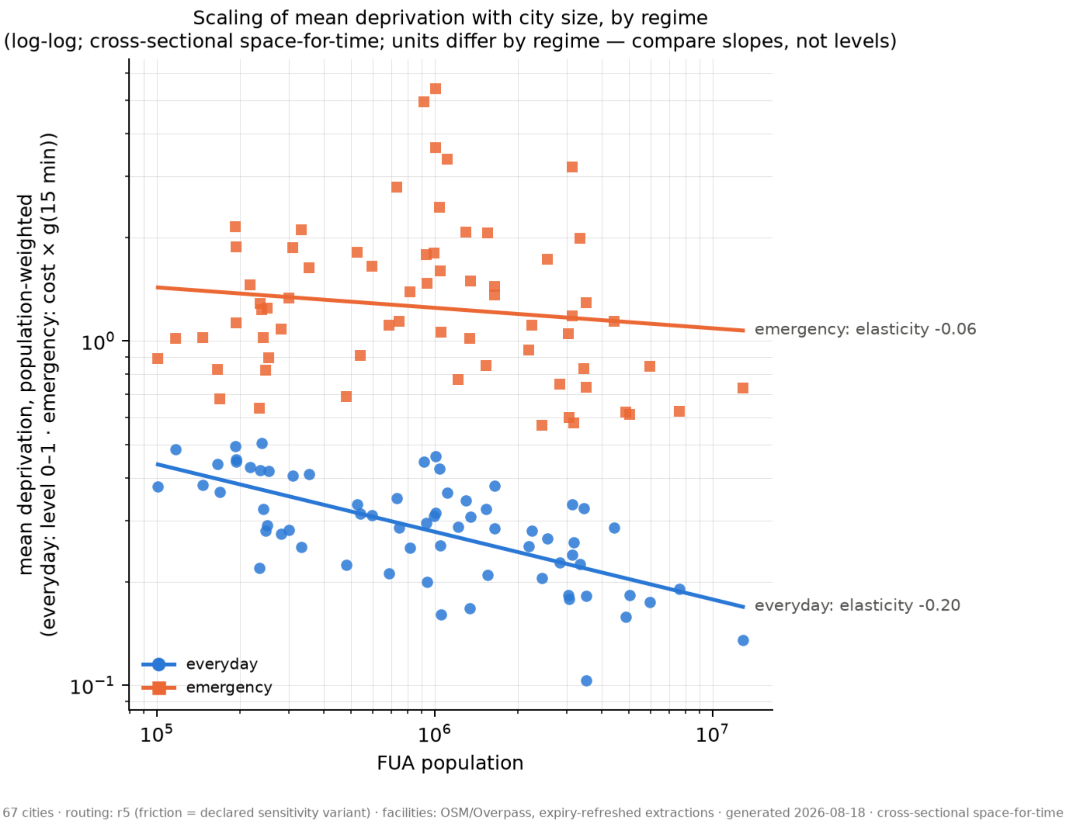

Figure 3: Regime-specific agglomeration. Population-weighted mean deprivation against FUA population (log-log) for both regimes across the 67 cities; the two series are in different units (everyday: level 0–1; emergency: cost in multiples of g(15)), so only the slopes compare, never the levels. Everyday deprivation level: elasticity -0.196, country-clustered wild bootstrap $p = 0.0002$, $R^2 = 0.44$. Emergency deprivation cost: elasticity -0.059, wild $p = 0.21$, bounded within ± 0.13 by equivalence testing. The paired slope difference of $+0.137$ is significant at wild $p = 0.002$. Gradients are cross-sectional (space-for-time).

population the way pharmacies do.

The two slopes also differ from each other, tested within cities rather than by comparing two separate regressions: a paired regime test (regime $\times$ log-population interaction, country-clustered) puts the everyday slope 0.137 below the emergency slope ($p_{\text{wild}} = 0.002$). RQ1's level half therefore has a clean answer: agglomeration benefits are regime-specific. Bigger cities deliver everyday services closer to people; proximity to emergency care does not improve detectably with size, and the difference between the regimes is itself significant. For scaling theory the implication is direct: the returns to scale that govern everyday provisioning do not extend across service classes, so no single exponent describes what city size does to service welfare.

2.2 Inequality scales with size only in the everyday regime

Within-city inequality answers RQ1's second half, and it inverts the comfort of the first. The Gini of everyday deprivation rises with population: elasticity $+0.062$ ($p_{\text{wild}} = 0.0012$). The emergency Gini is size-flat at the baseline parameterisation ($+0.004$, $p_{\text{wild}} = 0.84$), and the paired slope difference of -0.058 is significant over all 67 cities ($p_{\text{wild}} = 0.029$). Size therefore buys mean everyday access at the price of a wider spread around that mean, while emergency inequality is set elsewhere. The everyday half of the claim is defended by a specification curve [24] rather than by any single calibration (Fig. 4): re-estimating the slope under all 14 deprivation parameterisations (the curvature grid and the functional-form swaps of Section 3.1) keeps it positive throughout ($+0.026$ to $+0.075$) and significant in 13 of 14. The exception is the concave

Box-Cox everyday form swap (+0.026, $p = 0.083$), which we state rather than smooth over.

The emergency half is conditional, and we report the conditionality as a finding rather than resolve it by choice of parameterisation. The EMS benchmarks define the cost scale up to the 15-minute cut-off, and all three literature-grounded escalation forms are calibrated to coincide there. Beyond the cut-off no benchmark exists, and that is exactly where the forms diverge (Fig. 1, right panel). Under the bounded survival-based curve (1 = full deprivation, saturating around 30–45 minutes) the emergency Gini is size-flat (+0.001, $p = 0.97$). Under polynomial Box-Cox escalation, the published baseline, it is size-flat across the whole curvature sweep (−0.006 to +0.014, none significant), in agreement with the deprivation-free travel-time Gini, itself size-flat (−0.015, $p = 0.42$). Under exponential escalation the Gini rises with size (+0.098, $p = 0.0003$); the mechanism, verified in the data, is that exponential pricing turns the Gini into an index of extreme-tail depth (it saturates at a median of 0.96, decouples from the time distribution's inequality, $\rho = 0.04$, and tracks the beyond-45-minute population share, $\rho = 0.45$), and how many minutes past the cut-off the worst-served residents sit does grow with size, even though the tail's population share does not (beyond-30-minute share elasticity −0.003, $p = 0.99$). Two statements hold under every parameterisation: everyday inequality rises with size, and emergency inequality never falls with size. Whether emergency inequality is flat or rising is decided by the beyond-benchmark escalation form, which no clinical benchmark pins down; it is flat under two of the three hypotheses and rising under the third.

2.3 The geography of divergence: country-level structure and a coverage gradient

If size does not set emergency deprivation, what does? The divergence between regimes has a geography, and the honest unit for testing it is the country: macro-region is a property of countries, so whole countries are permuted across regions (Fig. 5). The emergency Gini differs by macro-region (Cohen $f = 0.74$, $p_{\text{perm}} = 0.0024$), as do the divergence gap ($p_{\text{perm}} = 0.010$), the everyday–emergency coupling ρ ($p_{\text{perm}} = 0.0015$) and both compounding measures ($p_{\text{perm}} = 0.0008$ for the HH share, $p_{\text{perm}} = 0.0023$ for the continuous intensity). Central-Eastern Europe compounds: the region pairs the highest emergency inequality (median emergency Gini 0.54) with the strongest coupling between the two surfaces (median $\rho = 0.55$), so the same populations tend to carry both deprivations. The West's inequality problem is everyday-side (median divergence gap −0.12); the North's is emergency-side (+0.09). The everyday Gini's apparent regional contrast does not survive country-level testing ($p_{\text{perm}} = 0.27$ despite a city-level ANOVA $p = 0.0007$), a cautionary illustration of why city-level inference over-claims here.

Do the 67 cities fall into types? No. k-means clustering on the full standardised city-vector separates 62 cities from 5 (silhouette 0.65, bootstrap ARI 0.87 [25, 26], Gaussian-null $p = 0.001$), and the isolated group is an outlier group rather than a typology: its agreement with the region partition is ARI = 0.001. Those five are Bucureşti, Rīga, Vilnius, Tallinn and Ljubljana, the emergency-desert capitals, separated by their beyond-threshold emergency travel-time tails. Ward clustering selects the same axis and cuts it more conservatively, isolating a nested subset of the two most extreme capitals (65/2). The nesting is evidence for a gradient: the two algorithms disagree about where to cut a severity ordering, and agree on the ordering itself. Re-clustering with the outliers removed finds one further split along the same axis at lower intensity: a 48/14 cut isolating partial-desert cities with real emergency tails (among them Oslo, Stockholm, Warszawa, Sofia, Zagreb, Palermo and Luxembourg; silhouette 0.38, bootstrap ARI 0.74, null $p = 0.001$, ARI vs region 0.07), inside which Ward again nests (56/6). A feature-set robustness check that re-runs the whole pipeline with the 8/15-minute benchmark shares added reproduces the main partition exactly (ARI 1.00) and the peeled cut at ARI 0.97, while flagging Brăila and Łomża, the two smallest CEE cities, as sitting on the covered/partial boundary; the same two cities sat on the covered side under the pre-refresh extraction state. The citable structure is therefore a one-dimensional *severity ordering* of emergency-periphery

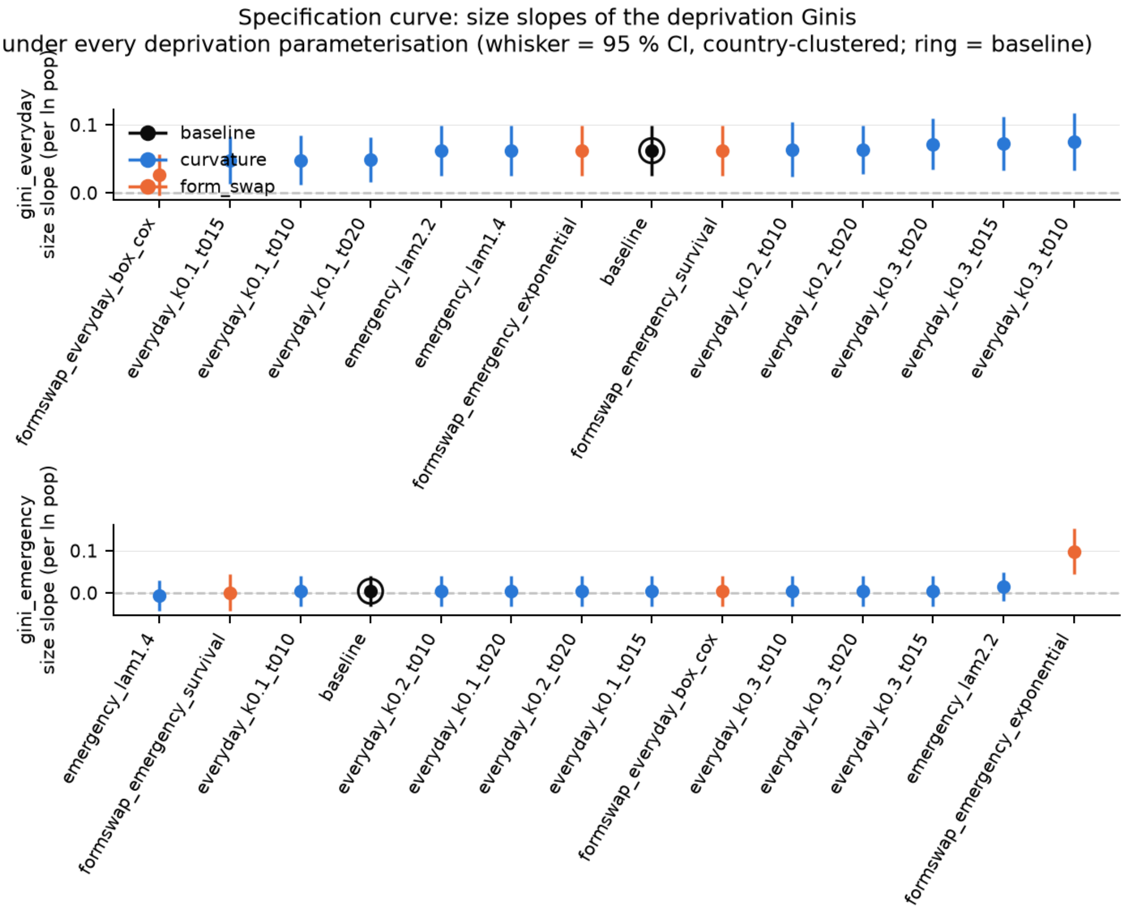

Figure 4: Specification curve of the inequality-scaling slopes. Top: everyday Gini; bottom: emergency Gini. Each point is the size elasticity of the within-city Gini re-estimated under the ringed baseline or one of the 14 alternative deprivation parameterisations (blue: curvature grid; orange: functional-form swaps); whiskers are country-clustered 95 % CIs. The everyday-Gini slope is positive under every specification (+0.026 to +0.075; significant in 13/14, the exception being the concave Box-Cox form swap at $p = 0.083$). The emergency-Gini slope splits by beyond-cut-off escalation form: flat under saturating and polynomial escalation, rising only under exponential escalation (+0.098, $p = 0.0003$).

coverage, covered (48) → partial desert (14) → desert (5), with a soft boundary between covered and partial and a robust ordering; we refer to these as coverage grades and never as city types.

The coverage grade, and no other variable we test, sets the emergency level. In a single country-clustered model with ln population centred at the sample mean, the grade dummies shift mean emergency deprivation overwhelmingly (Wald $p \approx 1.3 \times 10^{-15}$); the median mean cost rises 1.02 → 1.87 → 3.37 multiples of g(15) across covered → partial → desert (Fig. 6). The slope-by-grade interaction is $p = 0.010$ under the current 48/14/5 grades but was $p = 0.17$ under the pre-refresh 50/12/5 grades; a two-city boundary change flips it, so we treat the interaction as boundary-sensitive and do not headline any grade-specific size gradient. The level statement carries the content: coverage class, not city size, sets emergency deprivation. Consistent with this, the divergence-gap measures show at most a weak size gradient (gap slope −0.066 per $\log_{10}$ population, $p = 0.012$, $R^2 = 0.06$; Fig. S2), so size is a poor predictor of where the two regimes part ways (RQ2). What the deserts mean, and why access statistics cannot see them, is taken up in Section 2.5.

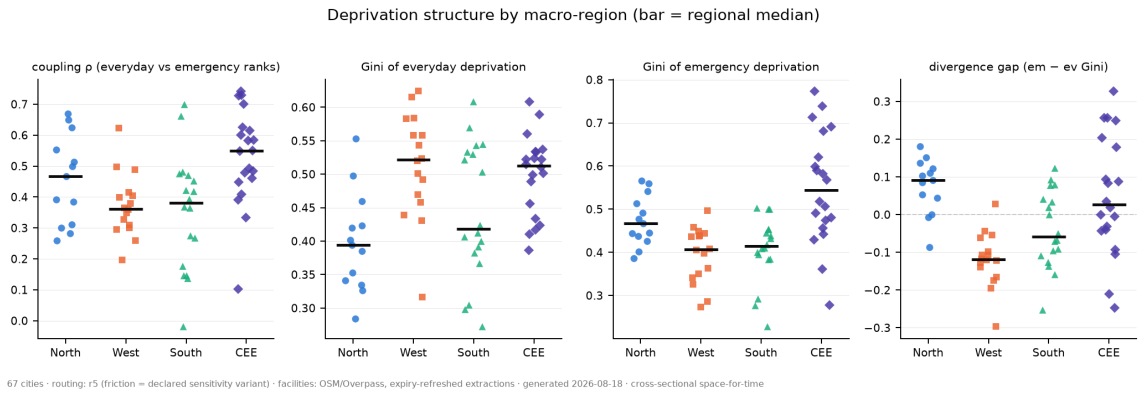

Figure 5: Regional structure of the divergence: strip plots by macro-region (bar = regional median) for the coupling ρ, the everyday Gini, the emergency Gini and the divergence gap. Contrasts that survive country-permutation testing: emergency Gini ($p_{\text{perm}} = 0.0024$), divergence gap ($p_{\text{perm}} = 0.010$), coupling ρ ($p_{\text{perm}} = 0.0015$); the compounding measures, tested in the text, also differ regionally (HH share $p_{\text{perm}} = 0.0008$; continuous intensity $p_{\text{perm}} = 0.0023$). The everyday Gini's apparent contrast does not survive country-level testing ($p_{\text{perm}} = 0.27$) and is shown without a claim. Region is a property of countries, so whole countries are permuted across regions; city-level p-values would be anti-conservative.

2.4 Compounding is the norm, and children carry it

How often do the same residents lose on both clocks (RQ3)? The co-location typology cuts both percentile surfaces at their population-weighted medians, which fixes the marginals at one half each; the whole 2×2 class table then has a single free number, the high-high (HH) compounding share, and statistical independence of the two surfaces corresponds to HH = 25 %. Everything above 25 % is excess co-location. The observed HH share averages 33 % and exceeds independence in 66 of 67 cities; the rank coupling ρ between the two surfaces spans -0.02 to 0.74 (Fig. 7). Compounding is not an exotic overlap of two rare conditions; across urban Europe it is the default relationship between the two deprivations.

Thresholded shares invite the question of the threshold, so the robustness harness prices it: the "how high is high" percentile cut moves the HH share by 29.8 percentage points at the median city, against 1.4 under the whole deprivation-curvature grid and 0.9 under the functional-form swaps, a roughly 21-fold dominance of the analyst's threshold over the welfare calibration. Two consequences follow. First, the headline compounding measure is the threshold-free continuous intensity, the population-weighted mean depth to which the typical resident sits inside *both* percentile distributions, with fixed anchors (1/3 under independence, 1/2 under perfect coupling, 1/4 under perfect divergence); it shows the same regional structure as the thresholded share ($p_{\text{perm}} = 0.0023$, strongest in Central-Eastern Europe). Second, every thresholded share in this paper travels with its threshold sweep. The spatial anatomy is consistent across contrasting cities (Fig. 8): compounding concentrates in peripheries where the saturated everyday level meets the escalating emergency cost, while dense cores hold most population in the low-low class. The maps are read under two rules stated once: legends carry population-weighted shares while the picture is area-weighted (a sparse periphery dominates visually at a modest population share), and no single cell's class is a finding (roughly 24 % of population flips class between routing engines); the per-city maps for all 67 cities are in the SI gallery.

Who lives in the compounded areas? Vulnerability enters on two strictly separated levels: EU-harmonised census age strata, comparable across all 67 cities and pooled; and national fine-grid strata (rent, tenure, finer ages), which are per-city depth and never pooled. Each stratum is compared against the cells where its own census variable is published (the covered-cell reference), because data coverage has its own geography. At the harmonised level the result is stark and one-sided: cells in the top quartile of child share carry more everyday deprivation than their

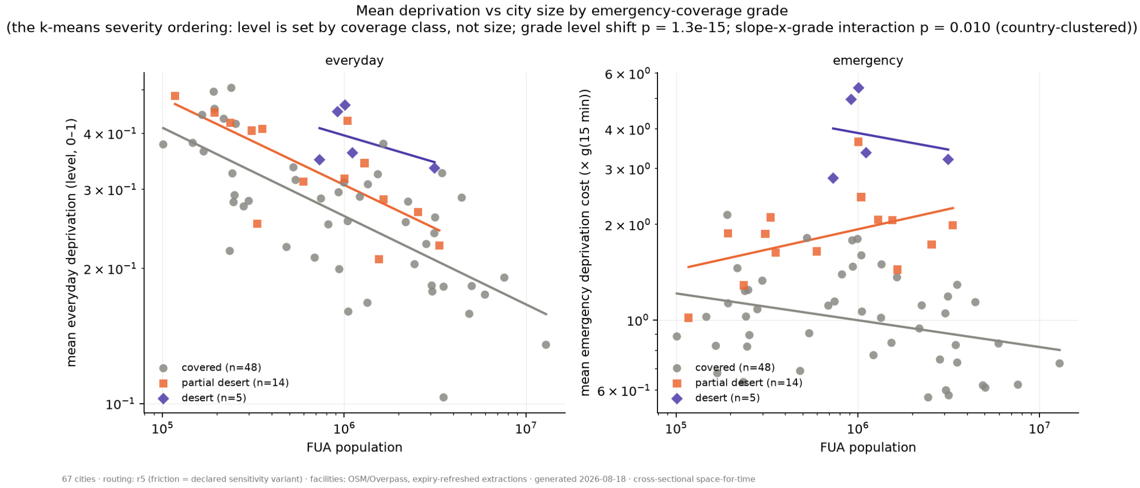

Figure 6: Mean deprivation against city size by emergency-coverage grade, per regime (left: everyday level; right: emergency cost, multiples of g(15)), grades read from the two clustering passes (covered 48 / partial desert 14 / desert 5). On the emergency side the grade shifts the level decisively (country-clustered Wald $p \approx 1.3 \times 10^{-15}$; median levels 1.02 / 1.87 / 3.37), while the slope-by-grade interaction is boundary-sensitive and not headlined. Within-grade fits are plain OLS over few cities, shown for orientation only.

comparison cells in 88 % of cities (median ratio 1.27) and more compounding in 88 % (median HH-share gap +0.13), strongest in Central-Eastern Europe (median ratio 1.44) and the North (1.30) (Fig. 9). The elderly show no such pattern at the European level: median ratio 0.96, above parity in only 37 % of cities, below parity across most of Central-Eastern Europe. That null is itself a result, and it is scale-specific: the national-data tier resolves sharper per-city patterns (in Hamburg, elderly 1.37, children 1.52, low-rent 1.61 against covered references), but those use different definitions and stay per-city. The European claim is about children; whatever sorting mechanism places families with children in the compounded periphery operates consistently across nearly the whole sample, in a group whose daily radius is short and whose emergency risk is real.

2.5 What deprivation adds over access

The framework's value has to be demonstrated against the state of the art, so every headline regression is re-estimated on the deprivation-free, minutes-based indicators the accessibility literature would use [5, 7], and the answer to RQ4 comes in three moves (Fig. 10; exact values in Table S1).

First, the claims survive dropping the deprivation layer entirely, which is what protects them from the suspicion of being calibration artefacts. The everyday size gradient is present in plain minutes (median walk time elasticity -0.237, $p = 3.8 \times 10^{-5}$; mean walk time -0.178), and the emergency non-gradient is present in minutes too (elasticities -0.043 to -0.059 across median, mean and p90 drive time). No headline of Sections 2.1–2.3 depends on the deprivation functions; the functions restate in welfare terms a structure the network geometry already carries. One detail of the minutes family already points at where that structure lives: the only minutes-based emergency statistic to reach conventional significance is the p90 drive time (-0.059, $p = 0.017$), the most tail-sensitive access statistic in the set. Even inside the access paradigm, whatever signal the emergency regime carries sits in the tail of the distribution rather than at its centre, which is precisely the region a convex cost function is built to price and an average over uniform minutes is built to dilute.

Second, the deprivation outcome is the better-behaved scientific object (Fig. 10A). City size explains 44 % of the variance in mean everyday deprivation against 16 % for the median

Table 2: The five emergency-desert capitals: access averages against deprivation. Median drive time to the nearest emergency facility and its ratio to the 67-city sample median (12 min) against mean emergency deprivation cost (multiples of g(15)) and its ratio to the sample value (1.14). Source: `desert_access_contrast.csv`.

City	Country	Median time (min)	Time ratio	Mean cost	Cost ratio
Vilnius	LT	14.0	1.17	5.39	4.72
Ljubljana	SI	23.0	1.92	4.96	4.34
Rīga	LV	13.5	1.13	3.37	2.95
Bucureşti	RO	8.5	0.71	3.21	2.81
Tallinn	EE	13.0	1.08	2.80	2.45

walk minute, its access counterpart, because the saturating level function discounts exactly the variation that carries no welfare content: whether a periphery cell sits 50 or 70 minutes from a pharmacy moves a minutes average and moves no household's daily reorganisation. Anchored deprivation acts as a welfare-weighted filter on travel time, and the filtered outcome is the one that scales cleanly. The filtering costs nothing in ordering: the deprivation-based and time-based Gini families agree closely in ranking cities (Pearson $r = 0.965$ everyday, 0.976 emergency), which is why every rank-based result in this paper is safe to call parameterisation-free, and why the level and inequality results are where the calibration carries information access measures cannot.

Third, and decisively, the emergency deserts of Section 2.3 are invisible to access averages (Fig. 10B; Table 2). Bucureşti's median emergency drive time is 8.5 minutes, 0.71 times the sample median: by the standard access statistic, the city looks better-served than typical urban Europe. Its mean emergency deprivation cost is 2.81 times the sample value. Across the five desert capitals, median-time ratios of 0.71–1.92 coexist with deprivation ratios of 2.45–4.72. The anatomy is always the same: a well-served core pulls the median minute down while a deep periphery tail, where drive times run far beyond the 15-minute benchmark, carries convex costs no average over uniform minutes can register. Only a loss function that prices the tail the way the clinical benchmarks say it should reveals the five capitals as the extreme of the coverage ordering; the same logic explains why the coverage grade, an object built from those tails, sets emergency deprivation levels overwhelmingly (Wald $p \approx 1.3 \times 10^{-15}$, Section 2.3) while remaining absent from any minutes-average ranking.

The three moves together give the framework its warrant. Where deprivation and access agree, the agreement certifies the access literature's rank-level findings at continental scale; where they disagree, the disagreement is confined to exactly the objects (levels, inequality magnitudes, tails) for which uniform minutes were never a welfare measure, and there the anchored curves earn their keep.

3 Discussion

3.1 How robust is the deprivation layer itself?

A paper whose contribution is a measurement layer must expect that layer to be attacked first, and the answer has two halves that both need stating. What is varied is drawn in Fig. 1: the Layer-1 curvature grid ($k \in \{0.1, 0.2, 0.3\}$ per minute $\times$ $t_0 \in \{10, 15, 20\}$ minutes for the everyday logistic; $\lambda \in \{1.4, 1.8, 2.2\}$ for the emergency Box-Cox) and the Layer-2 form swaps calibrated to the same domain anchors (a concave Box-Cox everyday level; exponential and bounded survival-based emergency escalations), all against the linear loss of a pure-access average.

How far the results move is in Table 3. Levels are not robust, and are not meant to be: curvature alone moves a typical city's everyday Gini by 0.24 Gini points (median envelope width across the 67 cities), which is exactly why the calibration carries information and why every slope claim is defended by the specification curve rather than by one parameterisation.

Table 3: Sensitivity envelopes of the deprivation layer: median min–max envelope width across the 67 cities, per sweep axis and target. Curvature moves the Ginis (the calibration carries information); no axis of the calibration moves the compounding share materially compared with the "how high is high" threshold, which dominates by a factor of about 21. The form-swap emergency-Gini envelope (0.649) spans the three beyond-cut-off escalation hypotheses and is a boundary-of-validity statement, reported with H2's conditionality in Section 2.2. Source: `deprivation_sensitivity_summary.csv`.

Sweep axis	Everyday Gini	Emergency Gini	HH share
Curvature (Layer 1)	0.244	0.170	0.014
Form swap (Layer 2)	0.035	0.649	0.009
"How high is high" threshold	–	–	0.298

Ranks and classes are robust: the same curvature grid moves the compounding share by 1.4 percentage points at the median, the form swaps by 0.9, against 29.8 for the "how high is high" threshold. City-level Gini orderings survive with two scoped exceptions: rank agreement with the baseline stays at $\rho \geq 0.90$ over the curvature grid and the everyday form swap, while both emergency escalation swaps reorder the emergency-Gini ranking (exponential $\rho = 0.19$; bounded survival $\rho = 0.49$) precisely because each changes what that Gini measures, a change of measurand reported as such rather than noise.

The wider inventory reads as one block. Deprivation-variant flip cells, the population whose typology class changes under any deprivation parameterisation, average 18.1 % of city population (range 5.2–53.0 %), with the HH class share moving 2.1 percentage points on average and 6.1 at most. The clustering survives its own feature-set choice (main partition ARI 1.00, peeled 0.97 with benchmark shares included). The routing-engine cross-check bounds the engine's leverage: switching engines moves travel-time levels by 34–314 % and flips the class of roughly 24 % of population per city, while aggregate class shares move at most 0.7 percentage points; per-cell map readings are therefore never findings, aggregate shares are. OSM completeness was benchmarked against national registries where available (Germany: pharmacies 0.976, hospitals 1.24) and the external INKAR benchmark passed. Fig. S1 assembles the architecture in one view. The reading discipline it supports ran through the whole paper: rank-based results are near-invariant to the deprivation parameterisation, and level and inequality results are where the calibration carries the information.

3.2 What the results change for theory

1. Urban scaling: returns to scale in welfare are regime-specific. Scaling theory's power of generalisation rests on exponents that hold across cities, systems and quantities, with infrastructure in a single sublinear universality class [1]; its recent refinements condition exponents on national context [2]. Our results add a different kind of condition: the exponent depends on the welfare regime of the service class, within one urban system and one harmonised measurement. Everyday welfare scales (-0.196, explaining 44 % of cross-city variance); emergency welfare does not (bounded within ± 0.13); and the two slopes differ within cities ($p_{\text{wild}} = 0.002$). A single infrastructure exponent is therefore not a sufficient statistic for what city size does to service welfare, and theories that read agglomeration as generalised provisioning gain overstate what growth delivers. The inequality result cuts deeper: everyday welfare inequality *rises* with size under every parameterisation tested, so the same agglomeration that lowers mean everyday deprivation widens its spread. Scaling theory has largely concerned means; these results argue that the scaling of within-city distributions, and specifically of welfare-weighted distributions, is where the next generation of urban scaling laws has to live.

2. Accessibility and urbanism: impedance must be welfare-anchored. For the accessibility literature the contribution is a measurement theorem demonstrated at continental scale.

Where results are rank-based, minutes and deprivation agree almost perfectly ($r \geq 0.965$ between the Gini families), which certifies the field's rank-level findings. Where results concern levels, inequality magnitudes or tails, uniform minutes are not a welfare measure, and the divergence is not noise: the desert capitals sit at 0.71–1.92 times the sample by access and at 2.45–4.72 times it by deprivation (Table 2), and the choice of measure decides whether an emergency-side size story exists at all. The 15-minute-city doctrine [3, 4] emerges recast: its threshold anchors the everyday regime well, and precisely because that regime saturates, a proximity doctrine is an everyday-services doctrine, silent about the escalating-cost regime and about the compounded populations where both fail. The coverage grades (covered, partial desert, desert) offer the tail-sensitive screening object that averages cannot supply, and the framework as a whole (forms transferred from an estimated literature, curvature calibrated to domain anchors, robustness of ranks separated from meaning of levels) travels to any domain that possesses anchors of comparable strength.

3. Resilience: two capabilities, coupled where it hurts. Resilience theory has moved toward defining resilience as equitable access to essential services [19, 20] and has documented that access equality and network resilience reinforce each other [21], while empirical work shows everyday vulnerability failing to predict crisis capability [22]. Our two-regime measurement gives that distinction quantitative form across an urban system: the everyday and emergency surfaces are distinct objects with different scaling laws and different geographies, so a city's resilience portfolio cannot be read from either alone. The compounding result identifies the pathological case the coupled-systems argument predicts: the same residents in the periphery of both systems, above statistical independence in 66 of 67 cities, strongest exactly where emergency inequality is highest (Central-Eastern Europe). Compounding intensity, with its fixed anchors under independence and perfect coupling, is offered as a general-purpose measure of that coupling for any pair of capability surfaces.

3.3 Implications for planning and policy

The practical reading is short, because the two regimes point at two different jurisdictions. Everyday inequality is a big-city problem: it rises with size under every parameterisation, so the cities gaining everyday access on average are the ones whose internal spread widens, and intra-urban service planning is the lever. Emergency deprivation is a national coverage problem: it is set by coverage grade rather than size, differs by country and macro-region, and its extreme cases are five capital cities whose peripheries no municipal instrument can serve alone. The severity ordering gives national emergency-system planners a screening variable that access averages do not provide. And the consistent carriers of compounded deprivation at the European level are families with children, in 88 % of the cities studied, which makes school- and family-anchored service planning a compounding remedy rather than a sectoral one.

3.4 Limitations

Every size gradient is cross-sectional; the space-for-time reading was flagged once and applies throughout. The analysis covers walking for everyday services and driving for emergency care; public transit is deliberately out of scope, and the everyday regime is walk-only by design. Census strata arrive on a 1 km grid broadcast to the 100 m analysis grid, so they are neighbourhood attributes; employment is unreported in the census grid for Germany and France, so no harmonised SES slope exists there by design. Per-cell typology classes carry a roughly 24 % routing-engine flip share and are read only as spatial patterns. Population shares beyond fixed time thresholds contain the finite-fill constant assigned to service-deprived cells and are footnoted wherever used. The emergency-side external benchmark (INKAR) is unavailable, and the German registry counts carry a verify-before-publication flag.

Table 4: Facilities routed to (medians across the 67 FUAs) and population-weighted median travel times to the nearest facilities of each service. Everyday services are reached on foot, emergency services by car. Source: `accessibility_by_service_pooled.csv`.

Regime	Service	Facilities per FUA	Median time (min)
Everyday	general practitioner	93	7.9
Everyday	pharmacy	228	4.9
Everyday	supermarket	325	4.7
Everyday	school	409	4.4
Everyday	green space	636	3.8
Emergency	emergency department	6	12.0
Emergency	ambulance station	6	12.0

Two OSM completeness specifics deserve statement rather than smoothing. Mapped GP density varies about 12-fold across countries in a way that reflects tagging conventions rather than health systems: the median FUA has 10.8 mapped GPs per 100k inhabitants, yet the country medians are 2.1 in Sweden, 2.3 in Portugal, 2.6 in Finland and 3.5 in Italy against 24.5 across the West, because primary care there sits in health centres OSM tags differently. Everyday deprivation levels are correspondingly overstated in those countries, so cross-country everyday *level* comparisons are not supported (one plausible reason the everyday-Gini regional contrast fails country-level testing). The size gradients are unaffected: mapped GP density is uncorrelated with city size (log-log slope +0.09, country-clustered $p = 0.29$), and every scaling claim is country-clustered. Separately, eight cities have only one of the two emergency services mapped (Athina, Brăila, Helsinki, Lahti, Norrköping, Szeged, Talavera de la Reina, Žilina); the composite renormalises weights over the services present, their median mean emergency cost is 0.197 against 0.169 for the rest, the expected direction at a modest size, and none of them is in the desert group, so no headline turns on it. Table S3 reports the count per city.

4 Materials and Methods

Design and sample. Cross-sectional multi-city design with one harmonised city definition (Eurostat/OECD functional urban areas) and trajectories read from the cross-sectional size gradient [1, 2]; no longitudinal component exists, and all outputs are labelled space-for-time. The sample is a stratified draw over 4 macro-regions $\times$ 4 population strata: 67 FUAs, 24 countries, 113.9 million residents, 101,050 to 12.9 million inhabitants (Table 1; Table S3).

Grid, facilities, routing. The analysis grid is the GHS-POP 100 m population grid [27], cells clipped to each FUA. Facilities come from OpenStreetMap via Overpass queries with expiry-refreshed extractions; Table 4 summarises the routed facility sets. Travel times are computed with the R5 routing engine via r5py [28] over the OSM street network: walking for the everyday regime, driving for the emergency regime, with emergency matrices reverse-routed (station-to-cell, the substantively correct direction for ambulance response; measured car asymmetry median 1 minute). A friction-surface engine [6] is retained as a declared sensitivity variant after a two-city cross-check showed city-specific, uncorrectable level bias (E.1).

Everyday deprivation (a level). Per service, travel time is congestion-adjusted by a two-step floating catchment area (2SFCA) factor [29, 30]: facilities serving many people relative to capacity inflate the effective time of their users. A soft-minimum over reachable facilities then rewards substitutability (several similar options count as slightly closer than the nearest alone). The effective time passes through a logistic deprivation level function [10] with ceiling 1, inflection $t_0 = 15$ minutes (the 15-minute-city threshold [3]) and steepness $k = 0.2$ per minute;

the city surface is the weighted mean over the five service categories (school and green-space sub-types weigh 0.5 each). Cells with no network path to any facility are masked; routable cells beyond a service's cutoff receive a large finite time so they stay on the map at near-maximal deprivation.

Emergency deprivation (a cost). The emergency surface applies a convex Box-Cox deprivation cost function [11, 13] with $\lambda = 1.8$ (transferred; solving λ from the anchors gives 1.89–1.95, inside the sensitivity sweep) to the drive time to the nearest emergency facility. The anchors are the EMS response benchmarks: cost negligible below the 3–4 minute ideal [14, 17], about a third of the benchmark cost at the 8-minute urban standard (contested: no survival effect for most patients in [14], supported in [15]; attributed via [18], whose WHO 2020 reference is an unpinnable URL), and 1 at the 15-minute upper target [16, 18]; convexity is supported independently by distance-mortality evidence [31]. The 8-minute mark cannot carry a half-max anchor: convexity caps $g(8)/g(15)$ at 0.427 given $g(4)/g(15) = 0.10$, so the two regimes' anchor structures are genuinely different. All costs are reported in multiples of $g(15)$. Free-flow civilian drive times anchor the cost scale and are not literal response-time predictions (dispatch and turnout are unmodelled; lights-and-siren travel is faster than civilian flow; free-flow understates congested cores, which is conservative for the periphery findings).

Cross-regime standardisation. The everyday level and the emergency cost are on incomparable scales by design. Only unit-invariant statistics cross regimes: elasticities, Ginis (mean-normalised, the same move as comparing an income Gini with a wealth Gini) and rank-based measures. The co-location typology is built on population-weighted mid-rank percentile surfaces; at the median split the 2×2 class table has one free number, the HH share, and statistical independence corresponds to HH = 25 %. The continuous compounding intensity is the population-weighted mean of the smaller of the two percentile ranks (1/3 independence, 1/2 perfect coupling, 1/4 perfect divergence). The divergence gap (emergency Gini minus everyday Gini) is the one cross-regime subtraction, a difference of dimensionless indices, form-dependent, and never a standalone headline.

Equity strata. Census-2021 1 km strata (broad age groups) are broadcast to the grid and pooled across cities against covered-cell references (cells where the stratum's variable is published); national fine-grid strata (rent, tenure) are per-city depth and never pooled. Inequality within strata uses the concentration index [32].

Cross-city inference. Cities nest in 24 countries, so scaling regressions report country-clustered standard errors with wild cluster bootstrap p-values (null imposed) [33, 34]; regional contrasts are tested by permuting whole countries across macro-regions, with a country-random-intercept mixed model as triangulation; city-level ANOVA p-values are anti-conservative for such questions and are never cited as evidence. The emergency null is bounded by TOST equivalence tests [23]; slope robustness uses a specification curve over all 14 deprivation parameterisations [24]; clustering diagnostics report silhouette [26], bootstrap and cross-partition adjusted Rand indices [25] and a Gaussian silhouette null. Every headline elasticity carries leave-one-out, Cook's-distance, Huber and median-regression diagnostics.

Robustness harness. A structured (non-probabilistic) sweep recomputes rank-based targets over a defensible envelope: Layer 1 varies curvature within fixed forms; Layer 2 swaps the forms while pinning the same anchors (concave Box-Cox everyday; exponential and bounded survival-based emergency escalations); Layer 3 sweeps the accessibility model (soft-min κ, congestion exponent γ, bandwidth, k-nearest, mode set, unreachable policy); a threshold axis sweeps the typology cut. Results are reported as rank agreement, class-share envelopes, flip-cell shares and the specification curve (Section 3.1; Fig. S1).

Reproducibility. The pipeline is config-driven with provenance sidecars on every derived matrix; all tables and figures in this paper trace to the persisted 67-city state (every quoted number is checked against its source table by an automated audit of 133 checks run in CI). Code, configuration and the full result tables are available in the study repository.

Data availability. Population grid: GHS-POP R2023A [27]. Facilities and street networks: OpenStreetMap. Census strata: Eurostat Census 2021 grid. Per-city and cross-city result tables, per-city maps and sensitivity sweeps are persisted on the study repository's results branch.

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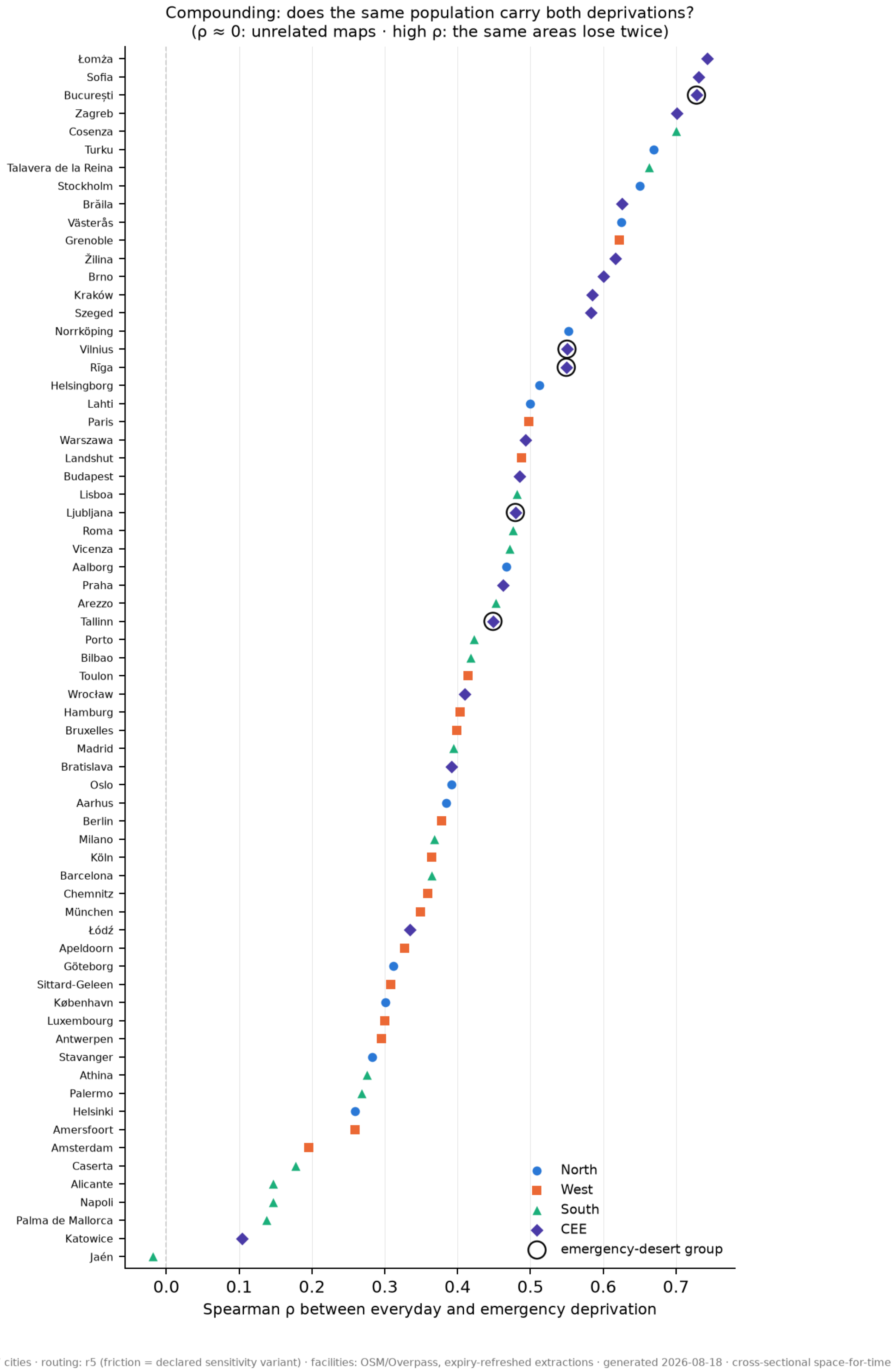

Figure 7: Coupling of the two deprivation surfaces across the 67 cities. Spearman ρ between the everyday and emergency population-weighted percentile surfaces, cities ranked; ρ spans -0.02 to 0.74. Colours mark macro-regions; the emergency-desert capitals are outlined. The HH compounding share exceeds the 25 % independence benchmark in 66 of 67 cities (sample mean 33 %). Rank-based measures are invariant to the deprivation parameterisation.

Figure 8: Compounding gallery: median-split co-location typology for contrasting cities, each panel labelled with the role it illustrates (all 67 maps in the SI). Colours: LL both low, HL high everyday only, LH high emergency only, HH compounding. Shares in the legends are population-weighted; the maps are area-weighted, so a sparse periphery dominates visually at a modest population share. Maps are read for spatial pattern only: roughly 24 % of population flips class between routing engines, so no per-cell class is a finding.

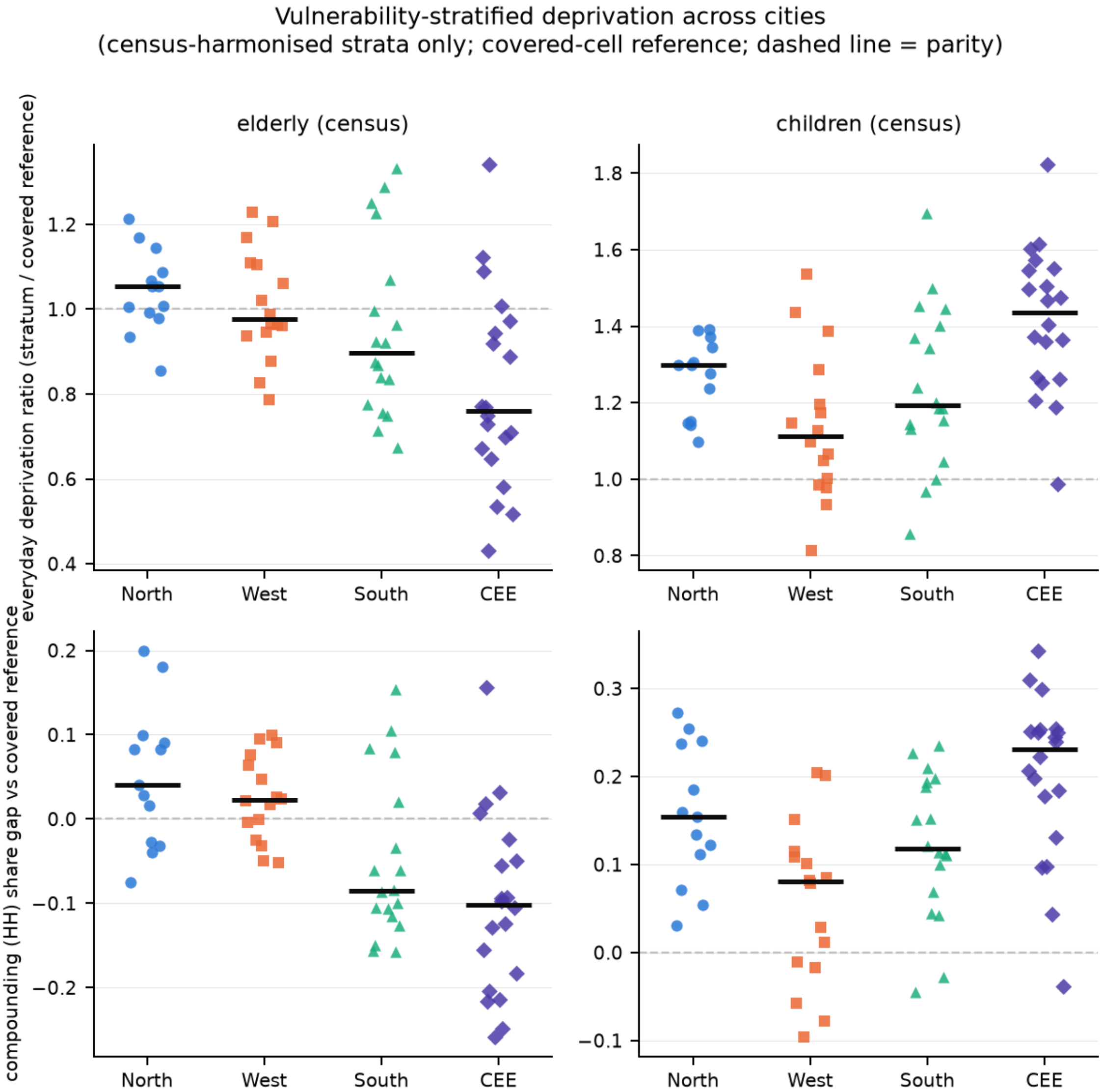

Figure 9: Who carries the deprivation: census-harmonised vulnerability strata by macro-region, pooled across the 67 cities against covered-cell references (dashed line = parity; bar = regional median). Columns: elderly and children strata; top row: everyday deprivation ratio (stratum vs reference); bottom row: compounding (HH) share gap. Children (top-quartile child-share cells) experience higher everyday deprivation in 88 % of cities (median ratio 1.27) and higher compounding in 88 % (median HH gap +0.13), strongest in CEE (1.44) and the North. The elderly stratum sits at parity (median 0.96, above 1 in 37 % of cities); no European elderly claim is made. National fine-grid strata are per-city depth and are never pooled with these.

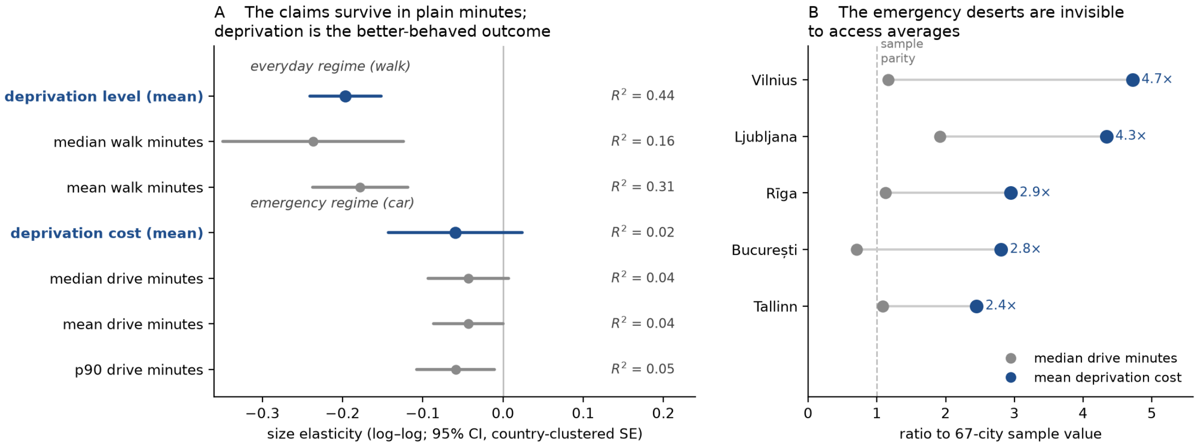

Figure 10: Deprivation against pure access. (A) Size elasticities of deprivation-based (blue) and minutes-based (grey) outcomes per regime (points: estimates; bars: 95 % CIs on country-clustered SEs; annotations: R^2). The everyday gradient and the emergency non-gradient are present in both families; the deprivation outcomes are the better-behaved ones. (B) The five emergency-desert capitals: ratio to the 67-city sample value under the access average (median drive minutes, grey) and under mean emergency deprivation cost (blue). All five sit near or below parity by access and at 2.45–4.72 times the sample by deprivation. Sources: `deprivation_vs_access.csv`, `desert_access_contrast.csv`.

Supplementary Information

The two geographies of urban deprivation

S1 The robustness architecture in one view

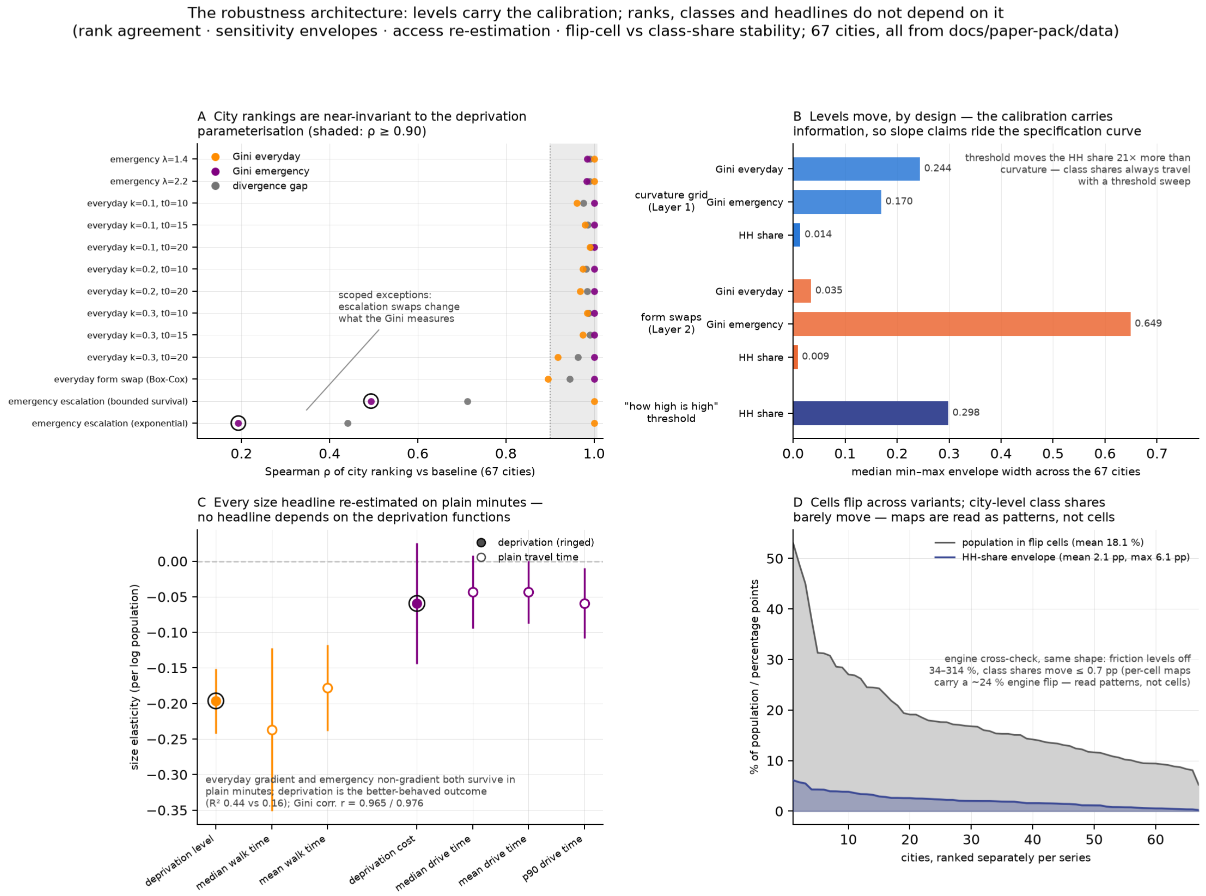

Figure S1: The robustness architecture (contribution claim 5). One view of the checks defending the results: rank agreement of city orderings across deprivation variants (min $\rho = 0.90$ over the curvature grid and everyday form swap; the two emergency escalation swaps are scoped exceptions that change the measurand, $\rho = 0.19$ exponential, 0.49 bounded survival); sensitivity envelope widths per axis (Table 3); access-based re-estimation of every headline (Table S1); and flip-cell against class-share stability (population-weighted flip share 18.1 % on average, 5.2–53.0 % across cities, against HH class-share movement of 2.1 pp average, 6.1 pp maximum), with the routing-engine cross-check quoted alongside (engine flip $\approx 24\,\%$, aggregate class shares ≤ 0.7 pp).

Rank-based results are near-invariant to the deprivation parameterisation; level and inequality results move with the calibration, which is where the calibration carries information. The flagged exceptions are exactly the two emergency escalation swaps, each of which changes what the emergency Gini measures (an extreme-tail-depth ordering under exponential escalation; a benchmark-window inequality ordering under the bounded survival curve), and they are reported as changes of measurand.

S2 Size gradient of the divergence measures

The companion gradients are weaker still: the coupling ρ ($p = 0.14$) and the compounding share ($p = 0.06$) show no significant size gradient (`size_gradient.csv`).

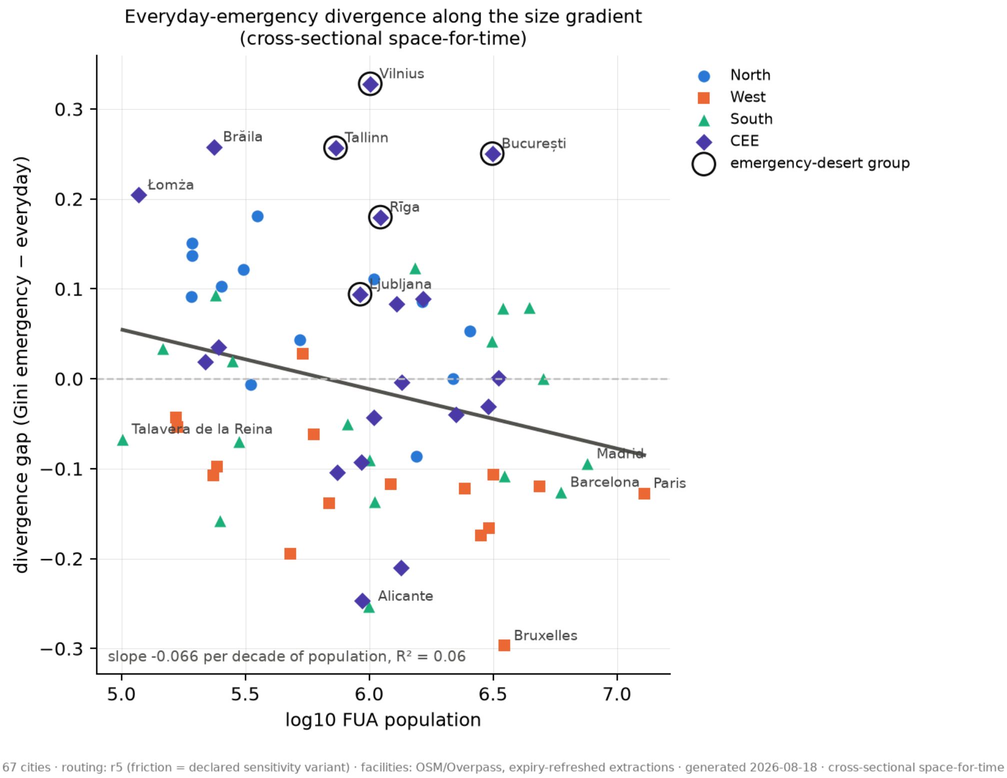

Figure S2: The divergence gap (emergency Gini minus everyday Gini) against $\log_{10}$ FUA population, coloured by macro-region with the emergency-desert capitals outlined. The gap falls weakly with size (-0.066 per decade of population, $p = 0.012$, $R^2 = 0.06$). Size is a poor predictor of where the two regimes diverge; the structure is regional and coverage-graded (main text Section 2.3).

S3 Deprivation vs access: exact values

Table S1: Size elasticities of deprivation-based and minutes-based outcomes (log-log, country-clustered SEs, 67 cities; cross-sectional space-for-time), plotted in main-text Fig. 10A. The everyday p90 is not regressed: above the walk cutoff the everyday composite time behaves as a count of unmet service categories rather than a travel time. Source: `deprivation_vs_access.csv`.

Regime	Outcome	Elasticity	p	R^2
Everyday	mean deprivation level	-0.196	< 0.001	0.44
Everyday	median walk time	-0.237	< 0.001	0.16
Everyday	mean walk time	-0.178	< 0.001	0.31
Emergency	mean deprivation cost	-0.059	0.16	0.02
Emergency	median drive time	-0.043	0.09	0.04
Emergency	mean drive time	-0.043	0.05	0.04
Emergency	p90 drive time	-0.059	0.02	0.05

S4 Validation exercises

E.1 Routing-engine cross-check. Re-running full cities under the friction-surface engine against the R5 baseline (Hamburg, Köln) shows travel-time levels understated by 34–314 % with city-specific Gini bias (emergency Gini -30 % in Hamburg, -18.5 % in Köln), which demoted friction to sensitivity variant. Aggregate typology class shares move ≤ 0.7 pp and the divergence

ρ by $\leq 11\,\%$, while roughly $24\,\%$ of population flips per-cell class; hence per-cell map readings are never findings.

E.2 OSM completeness. Country-level benchmark against national registries where available: Germany, pharmacies ratio 0.976, hospitals 1.24 (the hospital ratio carries a verify-before-publication flag on the registry side). The per-city facility counts (`accessibility_by_service_cities.csv`) provide the cross-country completeness picture the single-country registry benchmark cannot, and surface the GP tagging artefact discussed in the limitations.

E.3 External benchmark. The German INKAR accessibility indicators reproduce the study's German pharmacy and supermarket access patterns (passed); no comparable emergency-side benchmark is available.

E.4 Resolution and E.5 face validation. Quantile-curve comparisons of regime times under both engines and per-anchor-city face validation documents are maintained in the study repository.

Re-anchor verification. The emergency reporting anchor moved from $g(45)$ to $g(15)$ in August 2026. For the three cities re-seeded from stored artifacts the verification is exact: every rank-based indicator is identical to machine precision (max $|\Delta| \approx 10^{-16}$) and the level ratio equals $g(45)/g(15) = 6.7307$ to 10^{-15}; the other 64 cities were byte-identical between the two runs.

S5 Indicator glossary

Table S2: Compact glossary of the indicators used in this paper.

Term	Meaning
FUA	Functional urban area (Eurostat/OECD): city plus commuting zone; the unit "city".
Everyday regime	Five walkable service categories (GP, pharmacy, supermarket, school, green space); deprivation is a bounded *level* in $[0, 1]$.
Emergency regime	ED hospital and ambulance station, reached by car; deprivation is an unbounded convex *cost* in multiples of g(15).
$\bar{D}_{\mathrm{ev}}$, $\bar{C}_{\mathrm{em}}$	Population-weighted mean deprivation per regime; scales are incomparable across regimes.
G_{ev}, G_{em}	Population-weighted within-city Gini of each surface.
Divergence gap	$G_{\mathrm{em}} - G_{\mathrm{ev}}$; positive = emergency-side inequality problem.
ρ	Spearman correlation of the two population-weighted percentile surfaces.
LL/LH/HL/HH	Median-split co-location classes; HH = compounding. At the median split the table has one free number (HH); independence = 25 %.
Compounding intensity	Population-weighted mean of min(everyday pct, emergency pct); 1/3 independence, 1/2 coupled, 1/4 divergent.
Coverage grade	Severity ordering from the two clustering passes: covered / partial desert / desert.
Elasticity	Slope of log outcome on log population (space-for-time).
Wild cluster bootstrap	Country-clustered resampling giving reliable p-values with 24 clusters; the citable p for scaling claims.
p_{perm}	Permutation p-value reshuffling whole countries across macro-regions; the primary regional test.
TOST	Two one-sided equivalence tests; positive evidence a slope lies within a bound.
Specification curve	Headline regression re-estimated under every deprivation parameterisation.
Flip cells	Cells whose typology class changes under any deprivation variant.

S6 Per-city table

Table S3: The 67 functional urban areas, largest first. Grade: emergency-periphery coverage grade (Section 2.3); n_{em}: number of mapped emergency services (2 = both ED hospital and ambulance station); $\bar{D}_{\mathrm{ev}}$: population-weighted mean everyday deprivation level (0–1); $\bar{C}_{\mathrm{em}}$: mean emergency deprivation cost in multiples of the 15-min benchmark cost $g(15)$; G: population-weighted Gini per regime; ρ: Spearman coupling of the two percentile surfaces; HH: compounding population share at the median split.

City	CC	Region	Population	Grade	n_{em}	$\bar{D}_{\mathrm{ev}}$	$\bar{C}_{\mathrm{em}}$	G_{ev}	G_{em}	ρ	HH
Paris	FR	West	12,886,280	covered	2	0.135	0.73	0.625	0.497	0.50	0.334
Madrid	ES	South	7,601,394	covered	2	0.191	0.62	0.533	0.439	0.39	0.316
Barcelona	ES	South	5,945,709	covered	2	0.175	0.84	0.545	0.418	0.36	0.309
Milano	IT	South	5,030,024	covered	2	0.183	0.61	0.400	0.400	0.37	0.319
Berlin	DE	West	4,875,655	covered	2	0.158	0.62	0.558	0.438	0.38	0.317
Roma	IT	South	4,432,845	covered	2	0.287	1.14	0.424	0.503	0.48	0.342
Athina	EL	South	3,522,264	covered	1	0.103	0.73	0.609	0.500	0.28	0.274
Bruxelles	BE	West	3,521,497	covered	2	0.182	1.29	0.583	0.286	0.40	0.332
Napoli	IT	South	3,457,649	covered	2	0.326	0.83	0.305	0.384	0.15	0.270
Warszawa	PL	CEE	3,334,987	partial	2	0.225	1.99	0.589	0.590	0.49	0.347
Hamburg	DE	West	3,162,607	covered	2	0.260	0.58	0.543	0.437	0.40	0.321
Bucureşti	RO	CEE	3,140,502	desert	2	0.335	3.21	0.524	0.774	0.73	0.424
Lisboa	PT	South	3,132,904	covered	2	0.239	1.19	0.367	0.409	0.48	0.319
München	DE	West	3,054,507	covered	2	0.178	0.60	0.616	0.449	0.35	0.304
Budapest	HU	CEE	3,030,949	covered	2	0.183	1.05	0.538	0.507	0.48	0.327
Amsterdam	NL	West	2,829,893	covered	2	0.227	0.75	0.501	0.326	0.20	0.285
Stockholm	SE	North	2,549,795	partial	2	0.267	1.73	0.460	0.513	0.65	0.389
Köln	DE	West	2,438,338	covered	2	0.204	0.57	0.520	0.398	0.36	0.312
Praha	CZ	CEE	2,247,160	covered	2	0.282	1.11	0.515	0.475	0.46	0.354
København	DK	North	2,180,998	covered	2	0.254	0.94	0.401	0.401	0.30	0.307
Sofia	BG	CEE	1,647,936	partial	2	0.285	1.44	0.511	0.600	0.73	0.408
Helsinki	FI	North	1,642,879	covered	1	0.379	1.36	0.341	0.426	0.26	0.270
Oslo	NO	North	1,550,450	partial	2	0.209	2.06	0.553	0.467	0.39	0.297
Porto	PT	South	1,531,690	covered	2	0.324	0.85	0.272	0.395	0.42	0.321
Kraków	PL	CEE	1,350,286	covered	2	0.308	1.50	0.523	0.519	0.58	0.396
Katowice	PL	CEE	1,341,890	covered	2	0.168	1.02	0.489	0.279	0.10	0.267
Zagreb	HR	CEE	1,293,698	partial	2	0.343	2.07	0.499	0.583	0.70	0.395
Antwerpen	BE	West	1,219,366	covered	2	0.289	0.77	0.523	0.406	0.29	0.311
Rīga	LV	CEE	1,110,303	desert	2	0.362	3.37	0.502	0.682	0.55	0.374
Bilbao	ES	South	1,053,444	covered	2	0.160	1.06	0.570	0.433	0.42	0.324
Łódź	PL	CEE	1,045,886	covered	2	0.255	1.60	0.534	0.491	0.33	0.310
Göteborg	SE	North	1,042,720	partial	2	0.427	2.44	0.335	0.446	0.31	0.298
Vilnius	LT	CEE	1,010,376	desert	2	0.462	5.39	0.411	0.739	0.55	0.358
Palermo	IT	South	1,006,055	partial	2	0.317	3.64	0.383	0.292	0.27	0.289
Alicante	ES	South	996,867	covered	2	0.311	1.80	0.530	0.277	0.15	0.269
Wrocław	PL	CEE	939,076	covered	2	0.199	1.47	0.609	0.362	0.41	0.341
Bratislava	SK	CEE	931,613	covered	2	0.296	1.78	0.522	0.430	0.39	0.336
Ljubljana	SI	CEE	919,303	desert	2	0.447	4.96	0.387	0.481	0.48	0.341
Palma de Mallorca	ES	South	818,202	covered	2	0.251	1.39	0.503	0.453	0.14	0.294
Brno	CZ	CEE	745,397	covered	2	0.286	1.14	0.561	0.457	0.60	0.369
Tallinn	EE	CEE	731,883	desert	2	0.349	2.80	0.456	0.713	0.45	0.326
Grenoble	FR	West	688,170	covered	2	0.211	1.11	0.583	0.444	0.62	0.379
Luxembourg	LU	West	595,677	partial	2	0.312	1.64	0.469	0.407	0.30	0.301
Toulon	FR	West	540,279	covered	2	0.315	0.91	0.431	0.458	0.41	0.326
Aarhus	DK	North	525,545	covered	2	0.336	1.81	0.394	0.438	0.38	0.310
Chemnitz	DE	West	481,607	covered	2	0.223	0.69	0.558	0.363	0.36	0.319
Turku	FI	North	353,992	partial	2	0.409	1.63	0.385	0.565	0.67	0.360
Stavanger	NO	North	331,780	partial	2	0.252	2.10	0.498	0.492	0.28	0.299
Aalborg	DK	North	309,890	partial	2	0.405	1.87	0.419	0.541	0.47	0.342
Caserta	IT	South	298,402	covered	2	0.283	1.33	0.298	0.228	0.18	0.272
Vicenza	IT	South	280,054	covered	2	0.275	1.08	0.391	0.411	0.47	0.344
Helsingborg	SE	North	253,346	covered	2	0.420	0.90	0.284	0.387	0.51	0.366

Table S3 continued

City	CC	Region	Population	Grade	n_{em}	$\bar{D}_{\mathrm{ev}}$	$\bar{C}_{\mathrm{em}}$	G_{ev}	G_{em}	ρ	HH
Jaén	ES	South	249,477	covered	2	0.291	1.24	0.543	0.385	-0.02	0.233
Szeged	HU	CEE	246,224	covered	1	0.281	0.82	0.533	0.568	0.58	0.392
Apeldoorn	NL	West	242,638	covered	2	0.326	1.02	0.439	0.341	0.33	0.333
Cosenza	IT	South	239,305	covered	2	0.505	1.23	0.407	0.500	0.70	0.390
Brăila	RO	CEE	235,919	partial	1	0.422	1.29	0.434	0.692	0.63	0.359
Amersfoort	NL	West	234,650	covered	2	0.219	0.64	0.458	0.350	0.26	0.282
Žilina	SK	CEE	217,546	covered	1	0.431	1.46	0.424	0.443	0.62	0.365
Västerås	SE	North	193,510	covered	2	0.454	1.13	0.327	0.478	0.62	0.387
Norrköping	SE	North	193,317	partial	1	0.445	1.88	0.423	0.560	0.55	0.353
Lahti	FI	North	191,532	covered	1	0.495	2.14	0.353	0.444	0.50	0.345
Landshut	DE	West	168,617	covered	2	0.364	0.68	0.492	0.438	0.49	0.361
Sittard-Geleen	NL	West	165,544	covered	2	0.440	0.83	0.316	0.273	0.31	0.287
Arezzo	IT	South	146,788	covered	2	0.381	1.02	0.412	0.446	0.45	0.342
Łomża	PL	CEE	117,255	partial	2	0.484	1.01	0.417	0.621	0.74	0.430
Talavera de la Reina	ES	South	101,050	covered	1	0.378	0.89	0.522	0.454	0.66	0.415

S7 Per-city compounding maps

The median-split co-location typology map of each of the 67 cities, ordered by population. Legends carry population-weighted class shares; the maps are area-weighted, so coloured area and quoted share will not match by eye. Read for spatial pattern only: roughly 24 % of population flips class between routing engines (E.1), so no single cell's class is a finding.

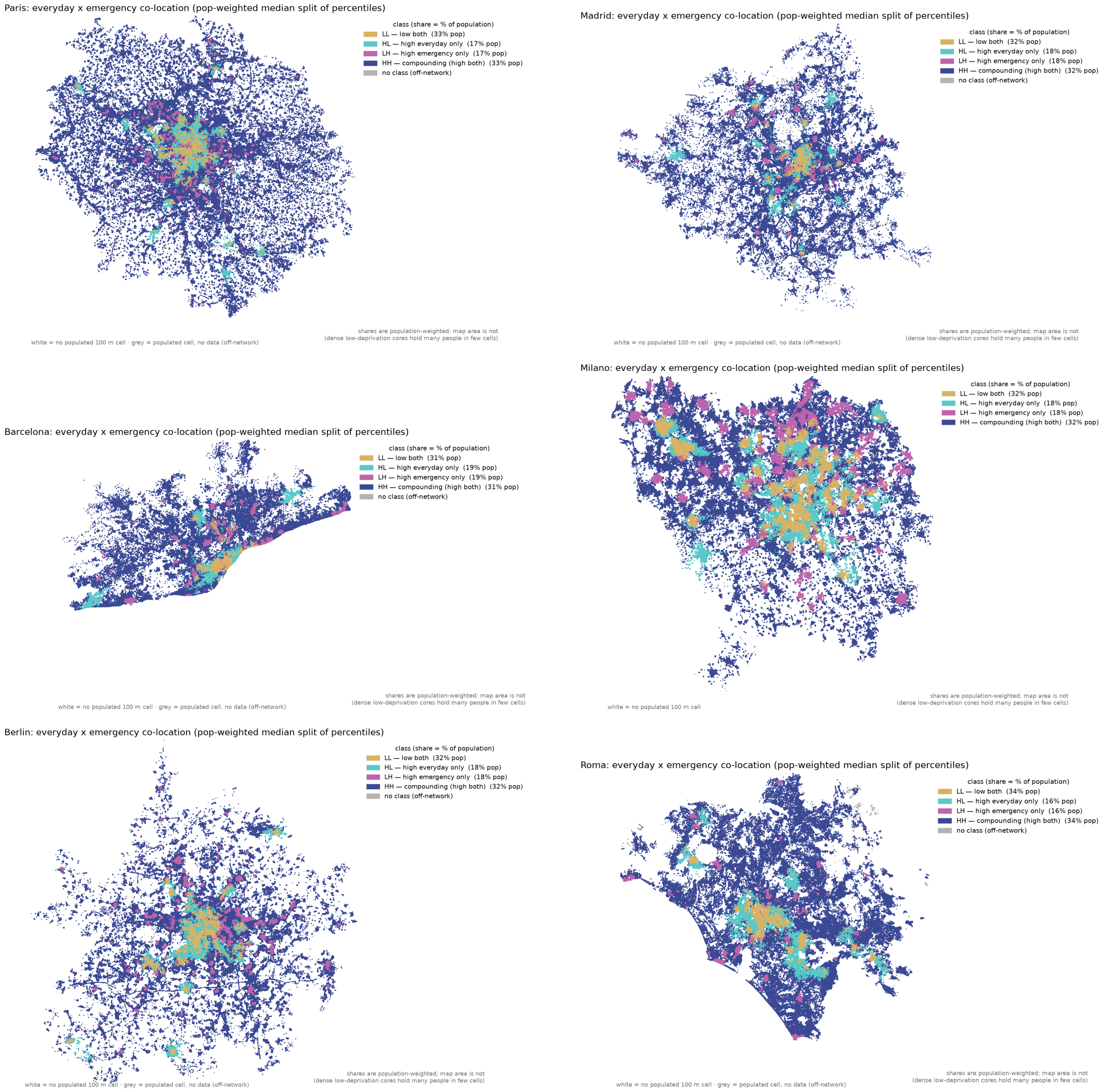

Figure S3: Per-city compounding maps (median-split co-location typology), cities 1–6 by population: Paris (FR), Madrid (ES), Barcelona (ES), Milano (IT), Berlin (DE), Roma (IT). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

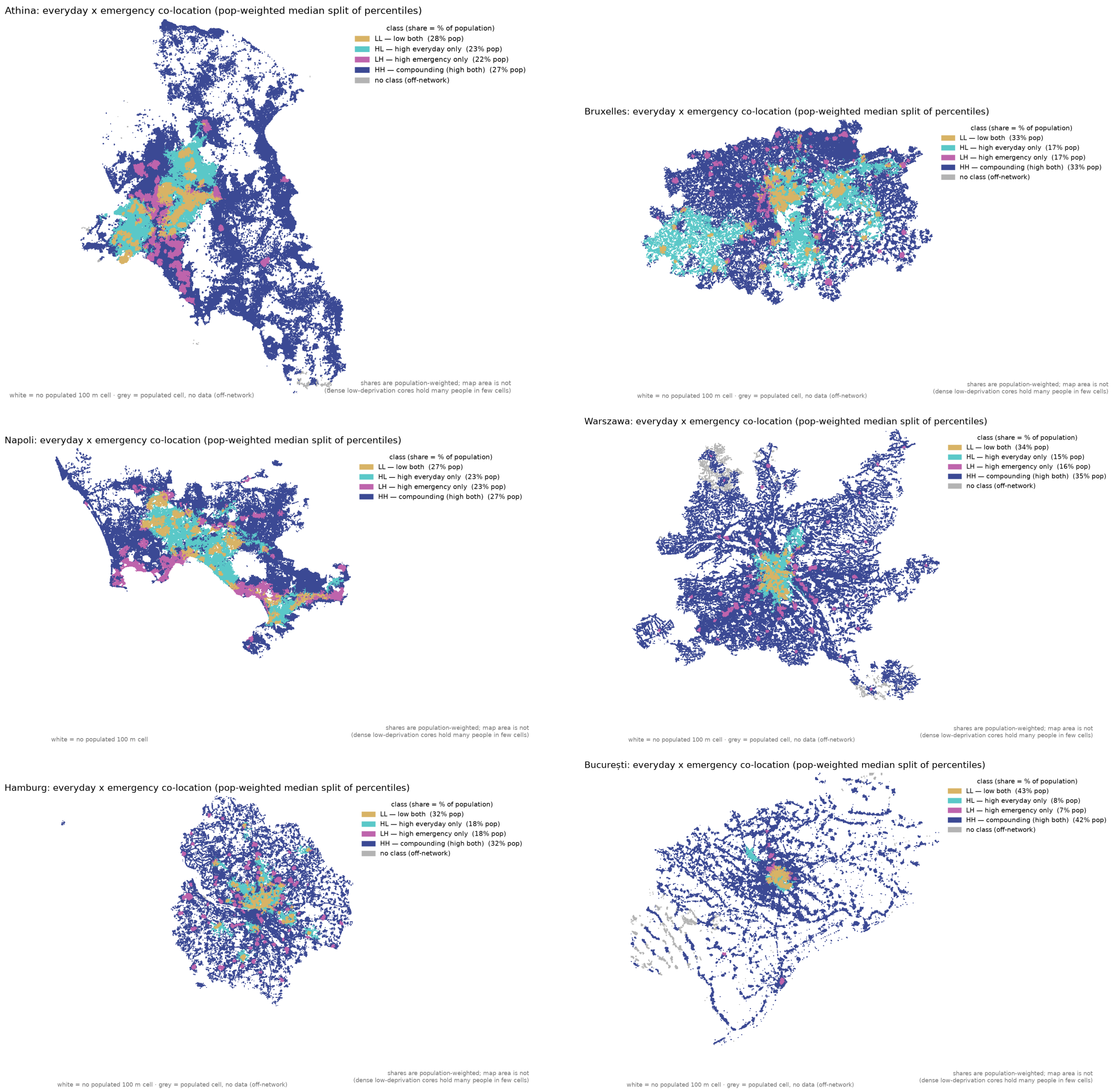

Figure S4: Per-city compounding maps (median-split co-location typology), cities 7–12 by population: Athina (EL), Bruxelles (BE), Napoli (IT), Warszawa (PL), Hamburg (DE), Bucureşti (RO). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

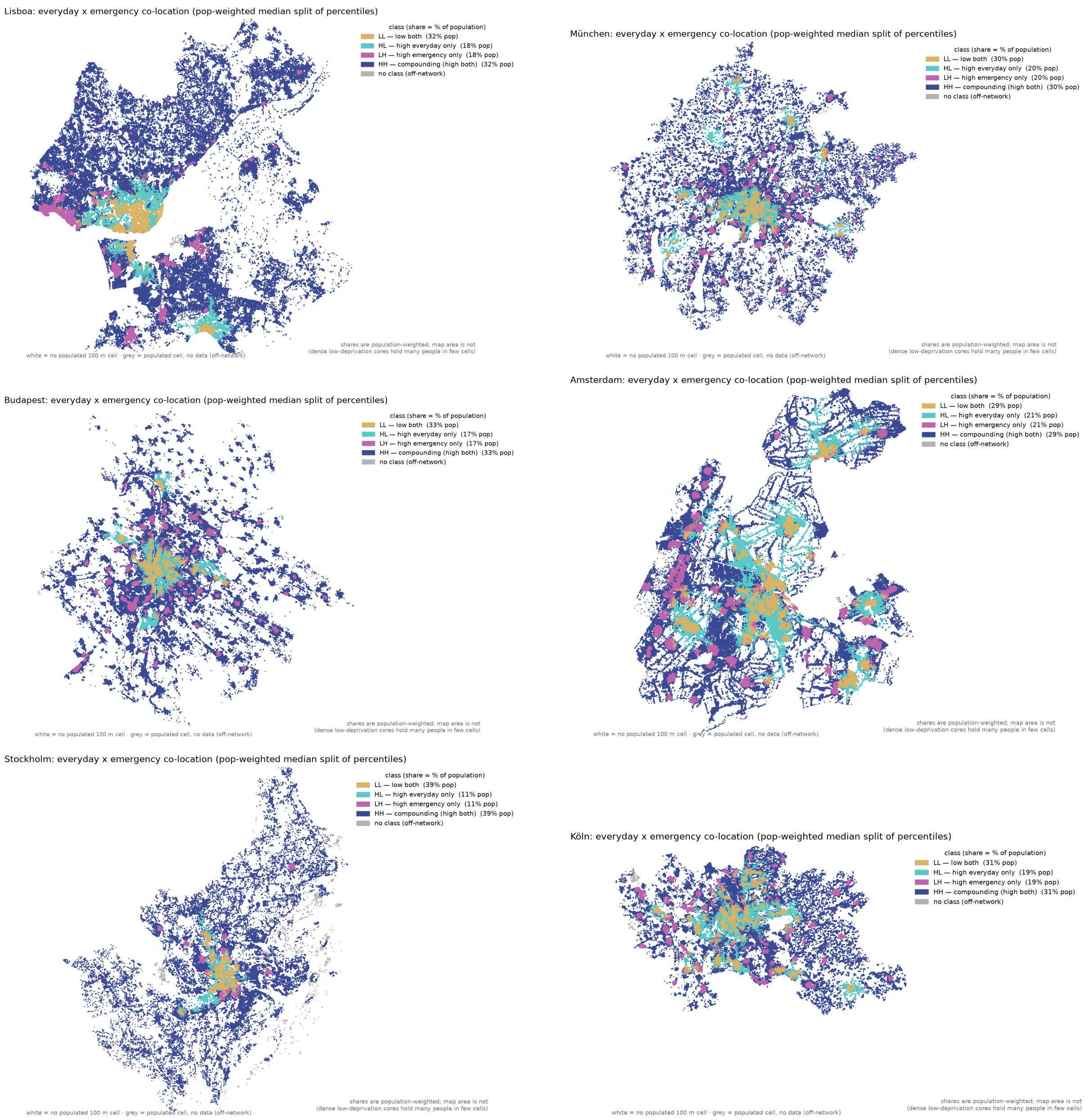

Figure S5: Per-city compounding maps (median-split co-location typology), cities 13–18 by population: Lisboa (PT), München (DE), Budapest (HU), Amsterdam (NL), Stockholm (SE), Köln (DE). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

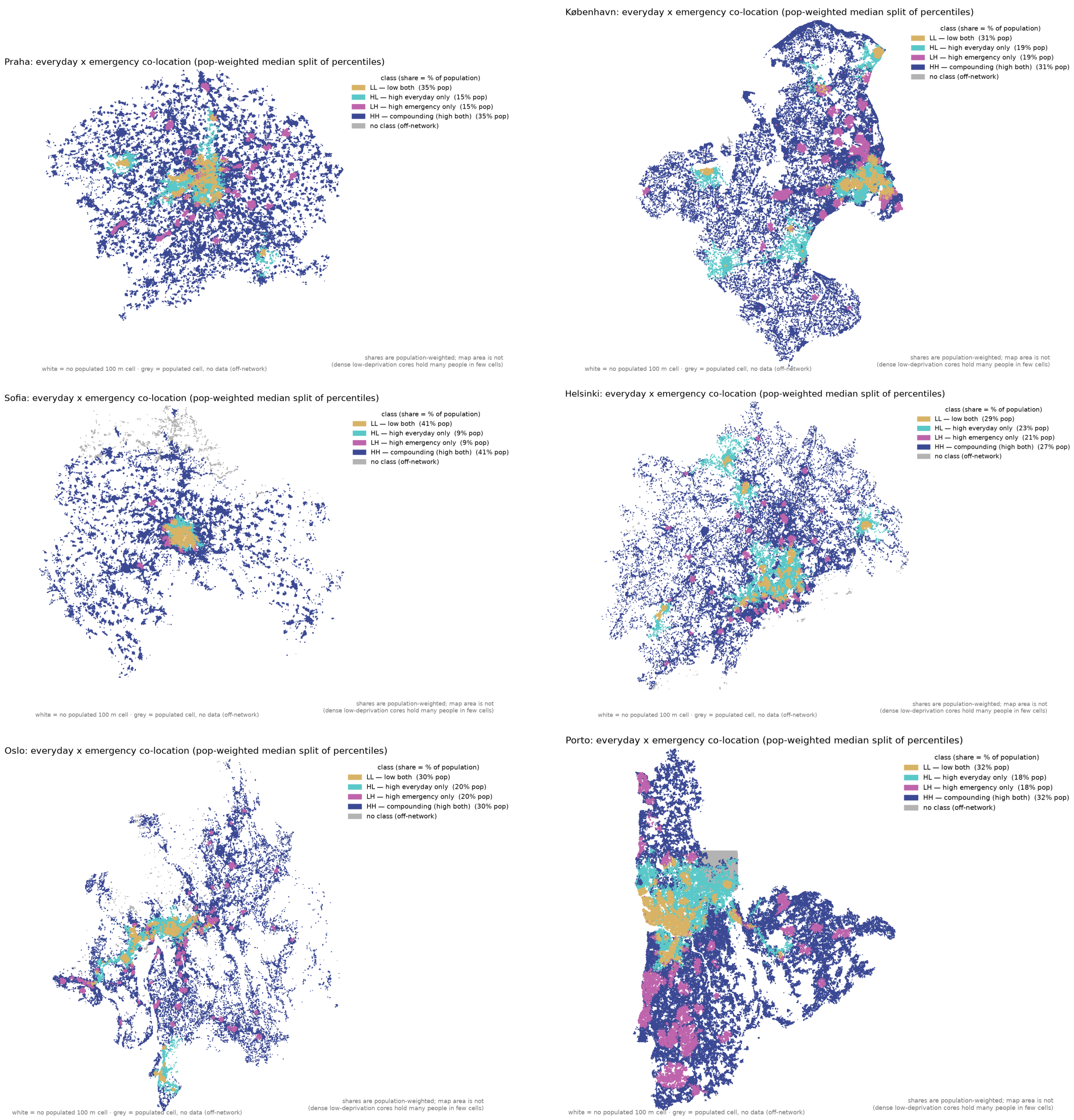

Figure S6: Per-city compounding maps (median-split co-location typology), cities 19–24 by population: Praha (CZ), København (DK), Sofia (BG), Helsinki (FI), Oslo (NO), Porto (PT). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

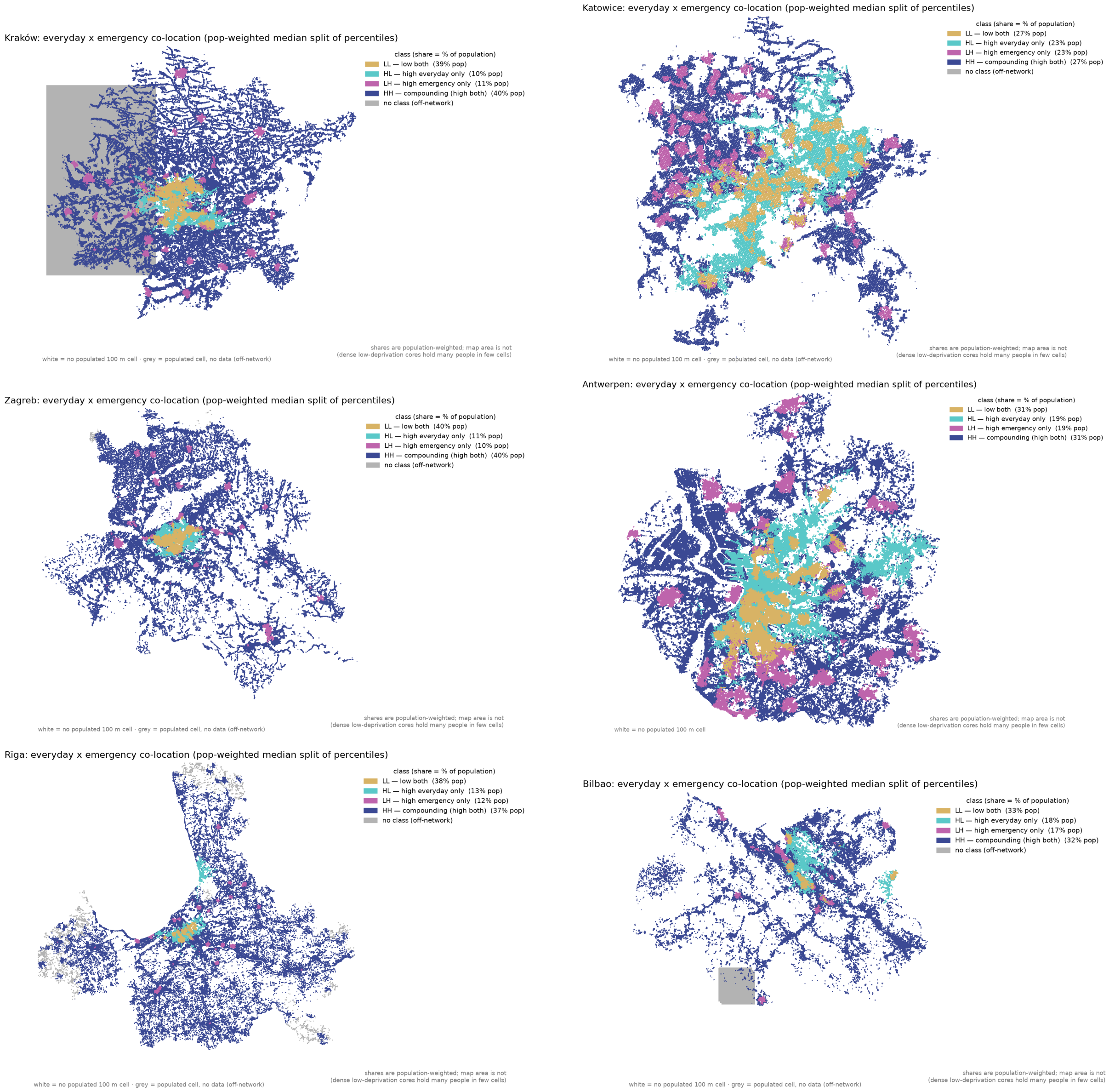

Figure S7: Per-city compounding maps (median-split co-location typology), cities 25–30 by population: Kraków (PL), Katowice (PL), Zagreb (HR), Antwerpen (BE), Rīga (LV), Bilbao (ES). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

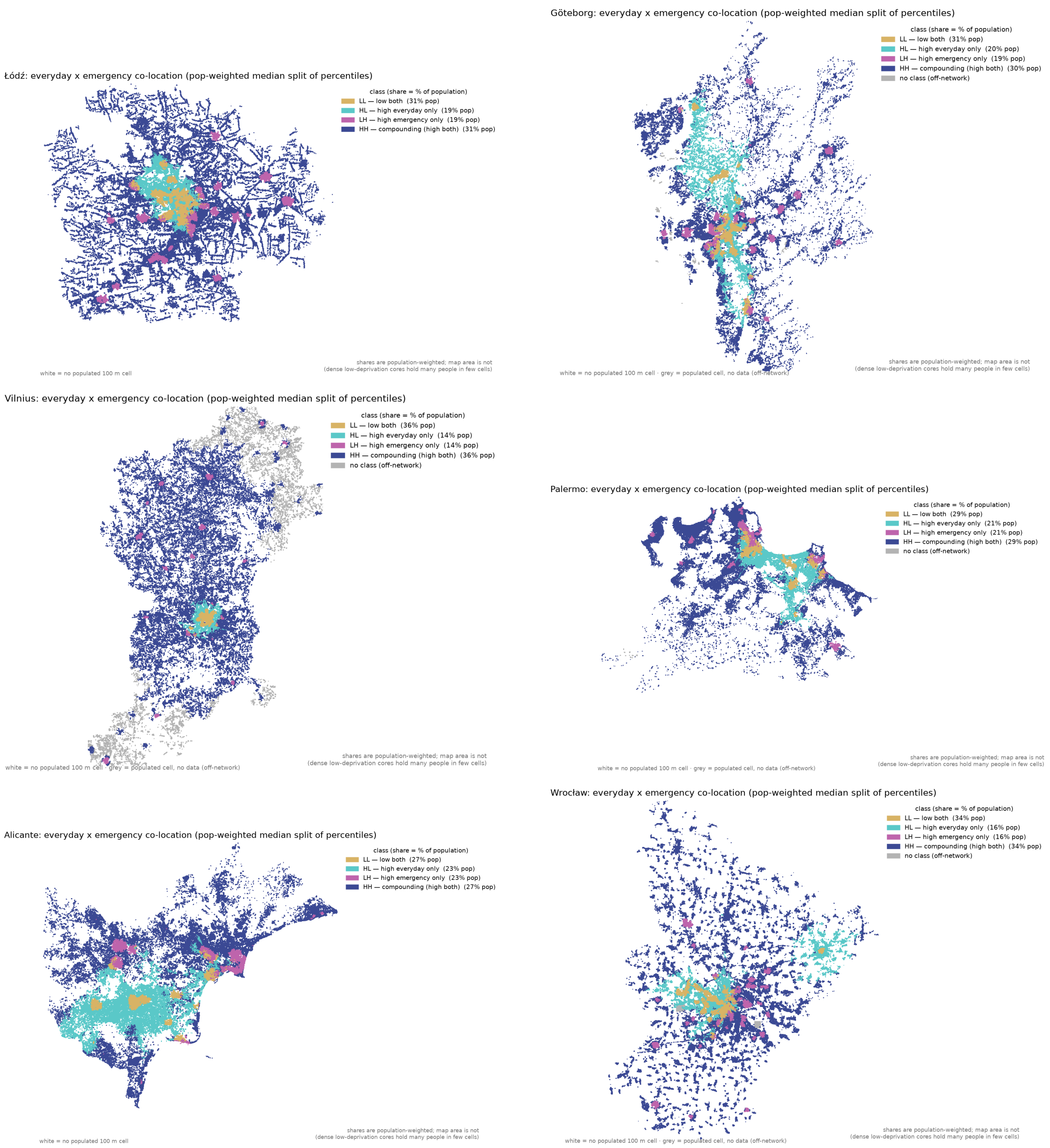

Figure S8: Per-city compounding maps (median-split co-location typology), cities 31–36 by population: Łódź (PL), Göteborg (SE), Vilnius (LT), Palermo (IT), Alicante (ES), Wrocław (PL). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

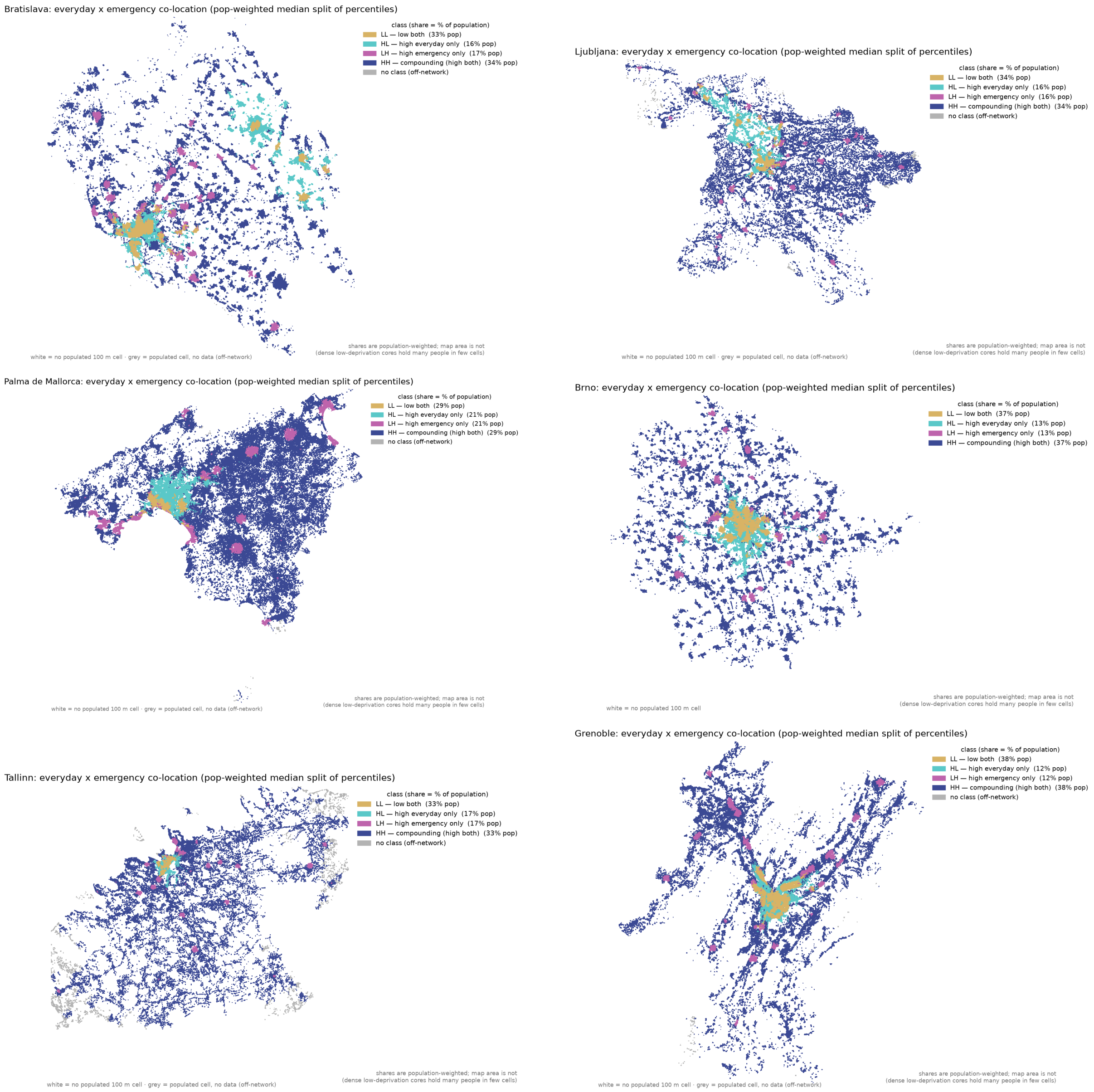

Figure S9: Per-city compounding maps (median-split co-location typology), cities 37–42 by population: Bratislava (SK), Ljubljana (SI), Palma de Mallorca (ES), Brno (CZ), Tallinn (EE), Grenoble (FR). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

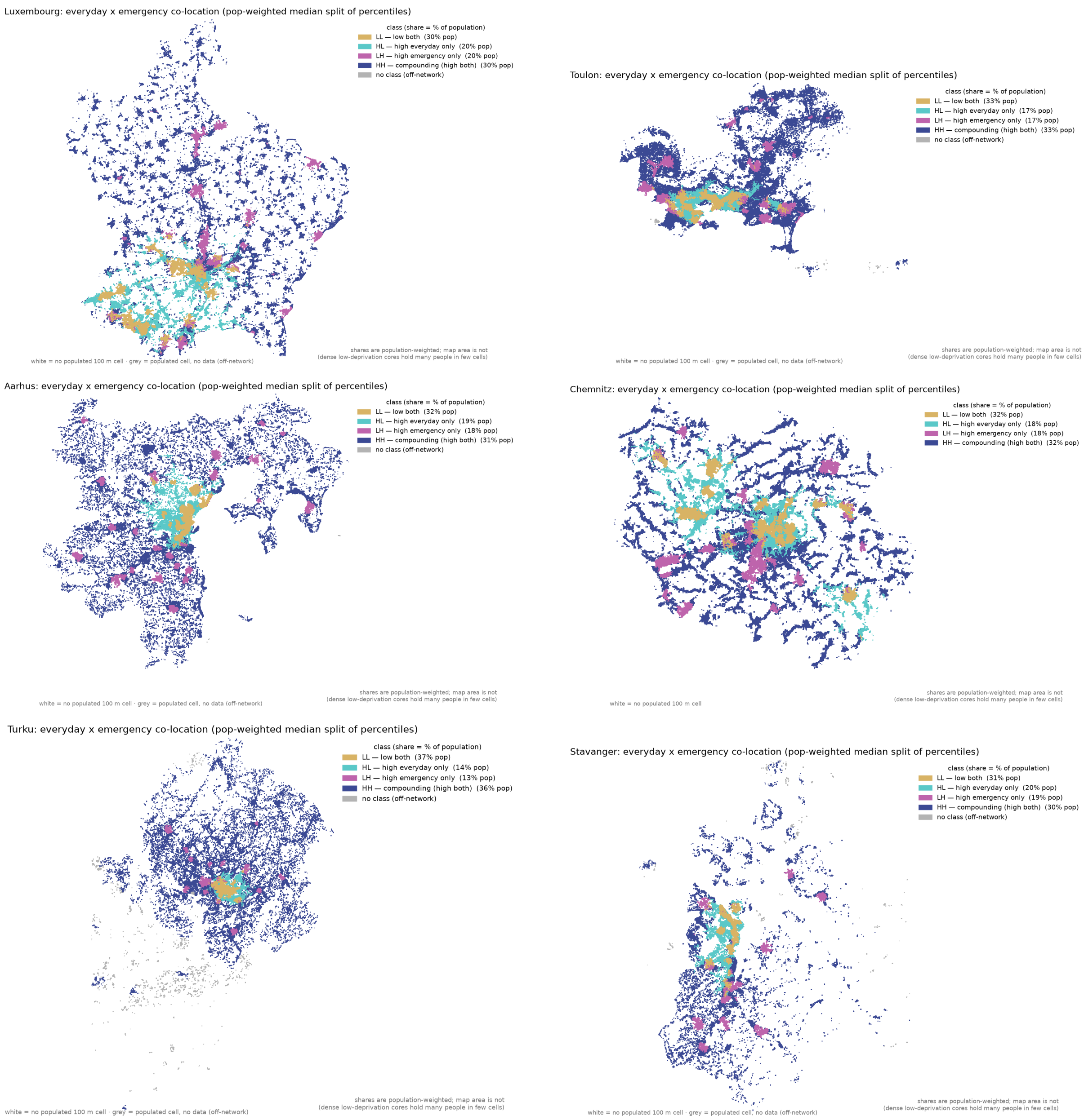

Figure S10: Per-city compounding maps (median-split co-location typology), cities 43–48 by population: Luxembourg (LU), Toulon (FR), Aarhus (DK), Chemnitz (DE), Turku (FI), Stavanger (NO). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

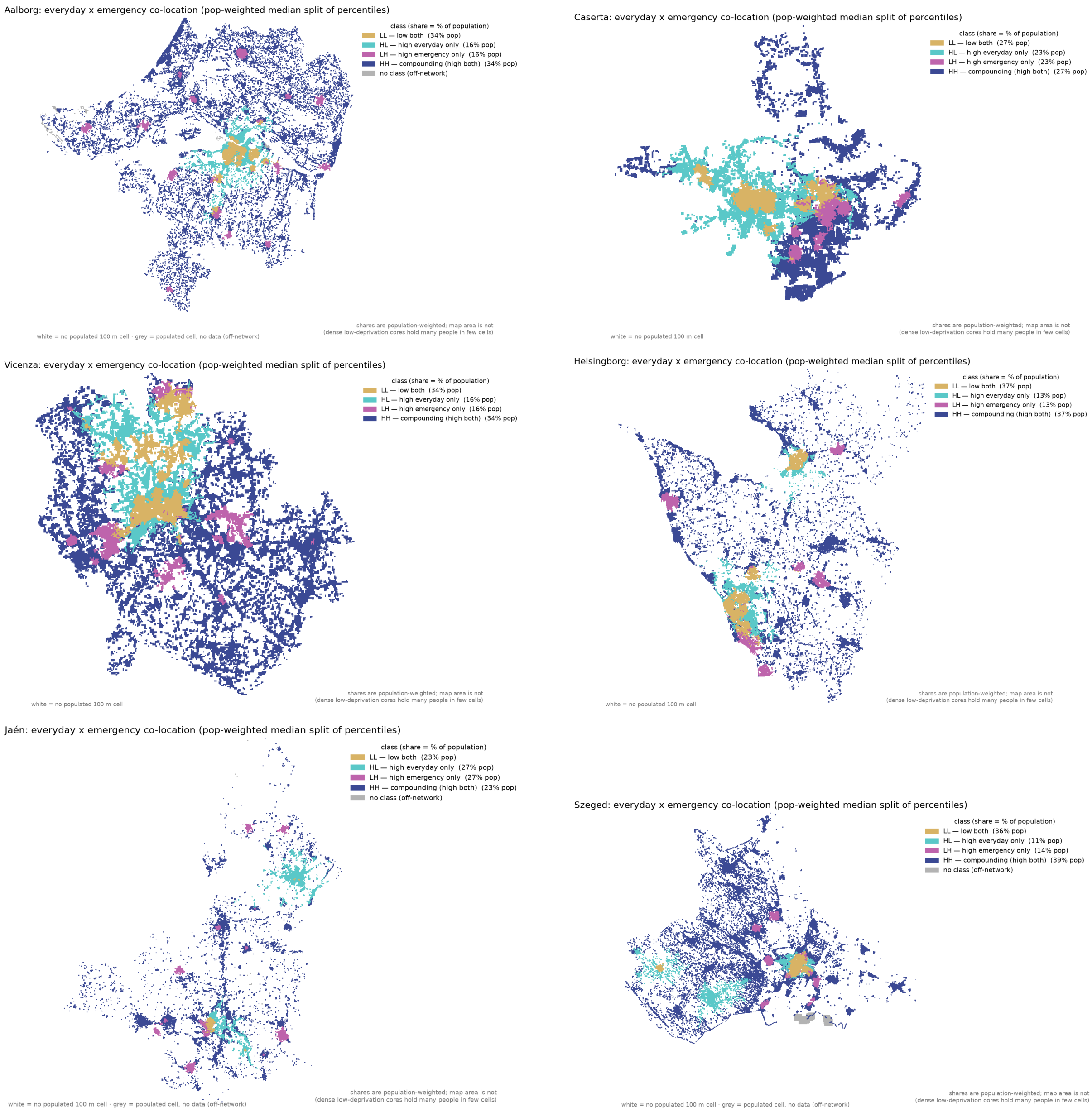

Figure S11: Per-city compounding maps (median-split co-location typology), cities 49–54 by population: Aalborg (DK), Caserta (IT), Vicenza (IT), Helsingborg (SE), Jaén (ES), Szeged (HU). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

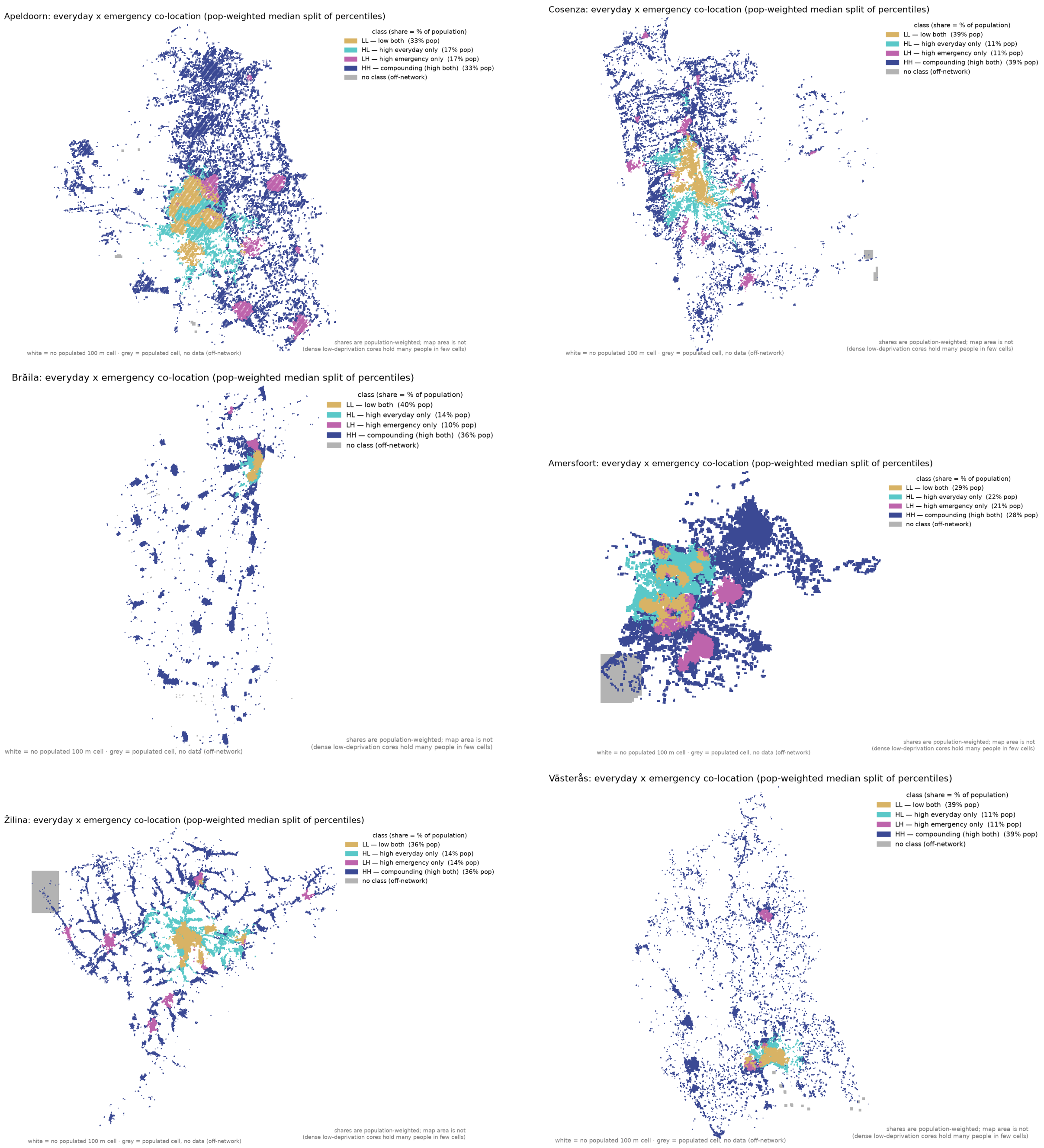

Figure S12: Per-city compounding maps (median-split co-location typology), cities 55–60 by population: Apeldoorn (NL), Cosenza (IT), Brăila (RO), Amersfoort (NL), Žilina (SK), Västerås (SE). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

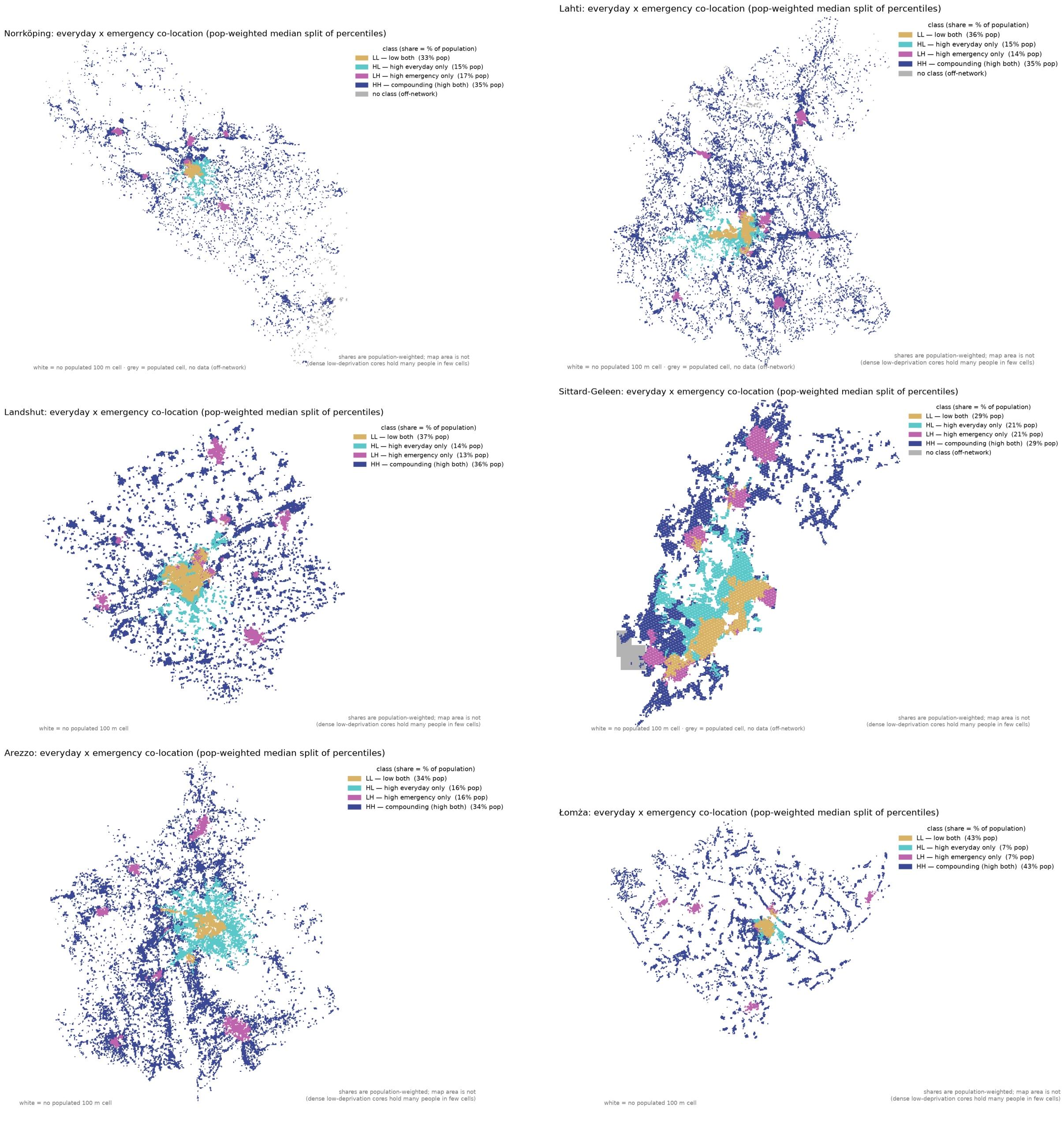

Figure S13: Per-city compounding maps (median-split co-location typology), cities 61–66 by population: Norrköping (SE), Lahti (FI), Landshut (DE), Sittard-Geleen (NL), Arezzo (IT), Łomża (PL). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

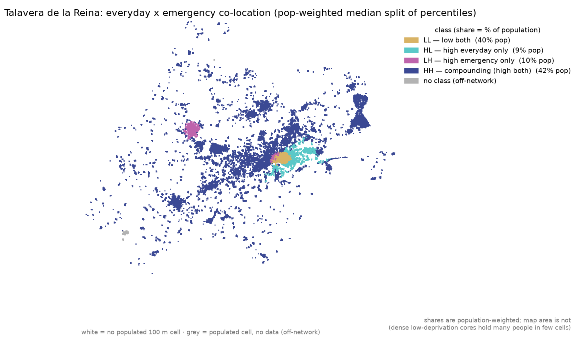

Figure S14: Per-city compounding maps (median-split co-location typology), cities 67–67 by population: Talavera de la Reina (ES). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).